

FoodAccessPREP

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2026-08-19

Background and Data

Our group used the 2015 and 2019 Food Access Research Atlas datasets, available through the U.S. Department of Agriculture Economic Research Service.

The datasets are of interest because they provide insight into access to supermarkets and essential food resources across different communities. They can be used to examine food access among people receiving benefits, those living in areas of high deprivation or low socioeconomic status, those living further away in urban vs rural areas, and different demographic groups. Using data from multiple years also allows us to identify changes over time and determine whether there have been noticeable improvements in food access.

The datasets contain information including census tract, poverty rate, food access at specified distances, and access measures across different racial and age groups, as well as SNAP beneficiaries. The majority of the variables are numerical, with some categorical variables such as census tract, state, and county. The data is structured so that individual variables represent specific measures of food access for particular demographic groups and distance thresholds.

One point of interest was that the 2019 dataset contained a large number of NULL values. These were assumed to represent zero values, as many of the affected variables represented whether an observation belonged to a particular demographic or access category. For example, a variable could represent the number of Native American individuals experiencing low food access at 20 miles, where a NULL indicated that there were no individuals meeting that condition. In contrast, both datasets also contained a considerable number of genuine NA values, particularly for `SNAP_la_rate` (2,405) and `senior_la_rate` (520).

To enable comparisons between years, our group used both the 2015 and 2019 Food Access Research Atlas datasets. The datasets were straightforward to integrate because they contained consistent and largely homogeneous variable names. We therefore retained the same variables from each dataset, allowing us to make meaningful comparisons in food access between 2015 and 2019.

Ethics, Privacy & Security

Ethics

This project involves data describing vulnerable populations, specifically seniors aged 65 and over and SNAP-receiving households. These populations require discretion as careless findings or framing about them can further stigmatise affected communities. A key consideration for this analysis is avoiding deficit oriented framing, practically, this means describing a tract as “low access” reflects a structural and geographic outcome specifically the distance for a household to supermarkets or transport infrastructure. It is not intended to signify the choices of the individuals living in the household, this is to avoid perpetuating any stereotypes or stigmas. As a result the language of the report has been mindfully selected to reflect this. For example we would state “A tract has a high rate of low-access seniors” instead of language that could be read as describing a senior or snap receiving households like “seniors in this area don’t shop at super markets”

Another ethical risk we are considering is an ecological fallacy. All analysis in this project is conducted at the census-tract level which uses aggregated rates for an area rather than individual data. So where a tract is classified as “low access” indicates the aggregate population meets USDA distance thresholds, it does not

mean that every individual within that tract lacks access. As a result our conclusions are based on patterns observed in tracts, not on individuals.

Privacy

This project uses only publicly available, government-released data (USDA Food Access Research Atlas and U.S. Census Bureau data), both provided at the census-tract level. We did no original data collection of our own, or have any interaction with the individuals involved with the data. The data is already aggregated tract-areas and released for public use as verified by the U.S. government so the privacy risk to individuals is minimal. The risks too to privacy are not technical with how the data is structured or analysed but in the interpretation of the data. There is still a risk (although very small risk) of individual identification for tracts with very small populations e.g a small amount of senior SNAP households in a rural tract, where in principle it could be easier to infer information about specific households. This represents a limitation of the dataset from a privacy perspective, though it is a low and indirect risk rather than a risk introduced by our analysis.

Security

As mentioned in the privacy section this project uses only public, pre-aggregated data. The overall security risk associated with this project is low. We practiced secure data management in this project where we store all our working data and code within a shared private Github repository, accessible only to the members of our group as opposed to a public repository or sharing code in any other external source. This gave us a controlled and access-limited location for our data, code and analysis. With some additional time and resources to allocate into security it would be worthwhile establishing a documented data deletion plan for after the project concludes.

DATA 2015

```
library(readxl)
DATA2015 <- read_excel("FoodAccessResearchAtlasData2015.xlsx",
                      sheet = 3, na = c("", "NULL"))
head(DATA2015)
```

A tibble: 6 x 147

##	CensusTract	State	County	Urban	POP2010	OHU2010	GroupQuartersFlag	NUMQTRS
##	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	01001020100	Alabama	Autauga	1	1912	693	0	0
## 2	01001020200	Alabama	Autauga	1	2170	743	0	181
## 3	01001020300	Alabama	Autauga	1	3373	1256	0	0
## 4	01001020400	Alabama	Autauga	1	4386	1722	0	0
## 5	01001020500	Alabama	Autauga	1	10766	4082	0	181
## 6	01001020600	Alabama	Autauga	1	3668	1311	0	0

i 139 more variables: PCTGQTRS <dbl>, LILATracts_1And10 <dbl>,
LILATracts_halfAnd10 <dbl>, LILATracts_1And20 <dbl>,
LILATracts_Vehicle <dbl>, HUNVFlag <dbl>, LowIncomeTracts <dbl>,
PovertyRate <dbl>, MedianFamilyIncome <dbl>, LA1and10 <dbl>,
LAhalfand10 <dbl>, LA1and20 <dbl>, LATracts_half <dbl>, LATracts1 <dbl>,
LATracts10 <dbl>, LATracts20 <dbl>, LATractsVehicle_20 <dbl>,
LAPOP1_10 <dbl>, LAPOP05_10 <dbl>, LAPOP1_20 <dbl>, LALOWI1_10 <dbl>, ...

DATA 2019

```
DATA2019 <- read_excel("FoodAccessResearchAtlasData2019.xlsx",
                      sheet = 3, na = c("", "NULL"))
head(DATA2019)
```

```
## # A tibble: 6 x 147
##   CensusTract State   County   Urban Pop2010 OHU2010 GroupQuartersFlag NUMGQTRS
##   <chr>         <chr>   <chr>   <dbl>   <dbl>   <dbl>         <dbl>   <dbl>
## 1 01001020100 Alabama Autauga ~     1    1912     693           0         0
## 2 01001020200 Alabama Autauga ~     1    2170     743           0        181
## 3 01001020300 Alabama Autauga ~     1    3373    1256           0         0
## 4 01001020400 Alabama Autauga ~     1    4386    1722           0         0
## 5 01001020500 Alabama Autauga ~     1   10766    4082           0        181
## 6 01001020600 Alabama Autauga ~     1    3668    1311           0         0
## # i 139 more variables: PCTGQTRS <dbl>, LILATracts_1And10 <dbl>,
## #   LILATracts_halfAnd10 <dbl>, LILATracts_1And20 <dbl>,
## #   LILATracts_Vehicle <dbl>, HUNVFlag <dbl>, LowIncomeTracts <dbl>,
## #   PovertyRate <dbl>, MedianFamilyIncome <dbl>, LA1and10 <dbl>,
## #   LAhalfand10 <dbl>, LA1and20 <dbl>, LATracts_half <dbl>, LATracts1 <dbl>,
## #   LATracts10 <dbl>, LATracts20 <dbl>, LATractsVehicle_20 <dbl>,
## #   LAPOP1_10 <dbl>, LAPOP05_10 <dbl>, LAPOP1_20 <dbl>, LALOWI1_10 <dbl>, ...
```

```
dim(DATA2015)
```

```
## [1] 72864   147
```

```
dim(DATA2019)
```

```
## [1] 72531   147
```

```
names(DATA2015)
```

```
##   [1] "CensusTract"      "State"          "County"
##   [4] "Urban"           "POP2010"        "OHU2010"
##   [7] "GroupQuartersFlag" "NUMGQTRS"       "PCTGQTRS"
##  [10] "LILATracts_1And10" "LILATracts_halfAnd10" "LILATracts_1And20"
##  [13] "LILATracts_Vehicle" "HUNVFlag"       "LowIncomeTracts"
##  [16] "PovertyRate"      "MedianFamilyIncome" "LA1and10"
##  [19] "LAhalfand10"      "LA1and20"       "LATracts_half"
##  [22] "LATracts1"        "LATracts10"     "LATracts20"
##  [25] "LATractsVehicle_20" "LAPOP1_10"      "LAPOP05_10"
##  [28] "LAPOP1_20"        "LALOWI1_10"     "LALOWI05_10"
##  [31] "LALOWI1_20"       "lapophalf"      "lapophalfshare"
##  [34] "lalowihalf"       "lalowihalfshare" "lakidshalf"
##  [37] "lakidshalfshare" "laseniorshalf"  "laseniorshalfshare"
##  [40] "lawwhitehalf"     "lawwhitehalfshare" "lablackhalf"
##  [43] "lablackhalfshare" "laasianhalf"    "laasianhalfshare"
##  [46] "lanhopihalf"      "lanhopihalfshare" "laaianhalf"
##  [49] "laaianhalfshare" "laomultirhalf"  "laomultirhalfshare"
##  [52] "lahisphalf"       "lahisphalfshare" "lahunvhalf"
##  [55] "lahunvhalfshare" "lasnaphalf"     "lasnaphalfshare"
##  [58] "lapop1"           "lapop1share"    "lalowi1"
##  [61] "lalowi1share"     "lakids1"        "lakids1share"
##  [64] "laseniors1"       "lablack1"       "lawwhite1"
##  [67] "lawwhite1share"   "lablack1share"  "lablack1share"
##  [70] "laasian1"         "laasian1share"  "lanhopi1"
##  [73] "lanhopi1share"    "laaian1"        "laaian1share"
##  [76] "laomultir1"       "laomultir1share" "lahisp1"
##  [79] "lahisp1share"     "lahunv1"        "lahunv1share"
##  [82] "lasnap1"          "lasnap1share"   "lapop10"
##  [85] "lapop10share"     "lalowi10"       "lalowi10share"
```

## [88]	"lakids10"	"lakids10share"	"laseniors10"
## [91]	"laseniors10share"	"lawwhite10"	"lawwhite10share"
## [94]	"lablack10"	"lablack10share"	"laasian10"
## [97]	"laasian10share"	"lanhopi10"	"lanhopi10share"
## [100]	"laaian10"	"laaian10share"	"laomultir10"
## [103]	"laomultir10share"	"lahisp10"	"lahisp10share"
## [106]	"lahunv10"	"lahunv10share"	"lasnap10"
## [109]	"lasnap10share"	"lapop20"	"lapop20share"
## [112]	"lalow120"	"lalow120share"	"lakids20"
## [115]	"lakids20share"	"laseniors20"	"laseniors20share"
## [118]	"lawwhite20"	"lawwhite20share"	"lablack20"
## [121]	"lablack20share"	"laasian20"	"laasian20share"
## [124]	"lanhopi20"	"lanhopi20share"	"laaian20"
## [127]	"laaian20share"	"laomultir20"	"laomultir20share"
## [130]	"lahisp20"	"lahisp20share"	"lahunv20"
## [133]	"lahunv20share"	"lasnap20"	"lasnap20share"
## [136]	"TractLOWI"	"TractKids"	"TractSeniors"
## [139]	"TractWhite"	"TractBlack"	"TractAsian"
## [142]	"TractNHOPI"	"TractAIAN"	"TractOMultir"
## [145]	"TractHispanic"	"TractHUNV"	"TractSNAP"

names(DATA2019)

## [1]	"CensusTract"	"State"	"County"
## [4]	"Urban"	"Pop2010"	"OHU2010"
## [7]	"GroupQuartersFlag"	"NUMGQTRS"	"PCTGQTRS"
## [10]	"LILATracts_1And10"	"LILATracts_halfAnd10"	"LILATracts_1And20"
## [13]	"LILATracts_Vehicle"	"HUNVFlag"	"LowIncomeTracts"
## [16]	"PovertyRate"	"MedianFamilyIncome"	"LA1and10"
## [19]	"LAhalfand10"	"LA1and20"	"LATracts_half"
## [22]	"LATracts1"	"LATracts10"	"LATracts20"
## [25]	"LATractsVehicle_20"	"LAPOP1_10"	"LAPOP05_10"
## [28]	"LAPOP1_20"	"LALOWI1_10"	"LALOWI05_10"
## [31]	"LALOWI1_20"	"lapophalf"	"lapophalfshare"
## [34]	"lalowihalf"	"lalowihalfshare"	"lakidshalf"
## [37]	"lakidshalfshare"	"laseniorshalf"	"laseniorshalfshare"
## [40]	"lawwhitehalf"	"lawwhitehalfshare"	"lablackhalf"
## [43]	"lablackhalfshare"	"laasianhalf"	"laasianhalfshare"
## [46]	"lanhopihalf"	"lanhopihalfshare"	"laaianhalf"
## [49]	"laaianhalfshare"	"laomultirhalf"	"laomultirhalfshare"
## [52]	"lahispalf"	"lahispalfshare"	"lahunvhalf"
## [55]	"lahunvhalfshare"	"lasnaphalf"	"lasnaphalfshare"
## [58]	"lapop1"	"lapop1share"	"lalow11"
## [61]	"lalow11share"	"lakids1"	"lakids1share"
## [64]	"laseniors1"	"laseniors1share"	"lawwhite1"
## [67]	"lawwhite1share"	"lablack1"	"lablack1share"
## [70]	"laasian1"	"laasian1share"	"lanhopi1"
## [73]	"lanhopi1share"	"laaian1"	"laaian1share"
## [76]	"laomultir1"	"laomultir1share"	"lahisp1"
## [79]	"lahisp1share"	"lahunv1"	"lahunv1share"
## [82]	"lasnap1"	"lasnap1share"	"lapop10"
## [85]	"lapop10share"	"lalow110"	"lalow110share"
## [88]	"lakids10"	"lakids10share"	"laseniors10"
## [91]	"laseniors10share"	"lawwhite10"	"lawwhite10share"
## [94]	"lablack10"	"lablack10share"	"laasian10"

```
## [97] "laasian10share"      "lanhopi10"          "lanhopi10share"
## [100] "laaian10"            "laaian10share"      "laomultir10"
## [103] "laomultir10share"    "lahisp10"           "lahisp10share"
## [106] "lahunv10"            "lahunv10share"      "lasnap10"
## [109] "lasnap10share"       "lapop20"            "lapop20share"
## [112] "lalow120"            "lalow120share"      "lakids20"
## [115] "lakids20share"       "laseniors20"         "laseniors20share"
## [118] "lawwhite20"          "lawwhite20share"     "lablack20"
## [121] "lablack20share"      "laasian20"           "laasian20share"
## [124] "lanhopi20"           "lanhopi20share"      "laaian20"
## [127] "laaian20share"       "laomultir20"         "laomultir20share"
## [130] "lahisp20"            "lahisp20share"      "lahunv20"
## [133] "lahunv20share"       "lasnap20"            "lasnap20share"
## [136] "TractLOWI"           "TractKids"           "TractSeniors"
## [139] "TractWhite"          "TractBlack"          "TractAsian"
## [142] "TractNHOPI"          "TractAIAN"           "TractOMultir"
## [145] "TractHispanic"       "TractHUNV"           "TractSNAP"
```

```
str(DATA2015)
```

```
## tibble [72,864 x 147] (S3: tbl_df/tbl/data.frame)
## $ CensusTract      : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
## $ State             : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
## $ County            : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
## $ Urban             : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
## $ POP2010           : num [1:72864] 1912 2170 3373 4386 10766 ...
## $ OHU2010           : num [1:72864] 693 743 1256 1722 4082 ...
## $ GroupQuartersFlag : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
## $ NUMGQTRS          : num [1:72864] 0 181 0 0 181 0 36 0 0 14 ...
## $ PCTGQTRS          : num [1:72864] 0 0.0834 0 0 0.0168 ...
## $ LILATracts_1And10 : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
## $ LILATracts_halfAnd10: num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
## $ LILATracts_1And20 : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
## $ LILATracts_Vehicle : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
## $ HUNVFlag          : num [1:72864] 0 0 0 0 1 0 1 1 0 0 ...
## $ LowIncomeTracts   : num [1:72864] 0 0 0 0 0 0 1 0 0 0 ...
## $ PovertyRate       : num [1:72864] 10 18.2 19.1 3.3 8.5 ...
## $ MedianFamilyIncome : num [1:72864] 74750 51875 52905 68079 77819 ...
## $ LA1and10          : num [1:72864] 1 0 1 1 1 1 1 0 0 1 ...
## $ LAhalfand10       : num [1:72864] 1 1 1 1 1 1 1 0 0 1 ...
## $ LA1and20          : num [1:72864] 1 0 1 1 1 1 1 0 0 0 ...
## $ LATracts_half     : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
## $ LATracts1         : num [1:72864] 1 0 1 1 1 1 1 0 0 0 ...
## $ LATracts10        : num [1:72864] 0 0 0 0 0 0 0 0 0 1 ...
## $ LATracts20        : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
## $ LATractsVehicle_20 : num [1:72864] 0 0 0 0 1 0 1 1 0 0 ...
## $ LAPOP1_10         : num [1:72864] 1357 483 1418 1363 2643 ...
## $ LAPOP05_10        : num [1:72864] 1732 1410 2765 3651 7778 ...
## $ LAPOP1_20         : num [1:72864] 1357 483 1418 1363 2643 ...
## $ LALOWI1_10        : num [1:72864] 322 145 697 410 623 ...
## $ LALOWI05_10       : num [1:72864] 412 475 1350 1068 1913 ...
## $ LALOWI1_20        : num [1:72864] 322 145 697 410 623 ...
## $ lapophalf         : num [1:72864] 1732 1410 2765 3651 7778 ...
## $ lapophalfshare    : num [1:72864] 0.906 0.65 0.82 0.832 0.722 ...
## $ lalowihalf        : num [1:72864] 412 475 1350 1068 1913 ...
```

```

## $ lalowihalfshare : num [1:72864] 0.215 0.219 0.4 0.243 0.178 ...
## $ lakidshalf : num [1:72864] 466 448 745 847 2309 ...
## $ lakidshalfshare : num [1:72864] 0.244 0.207 0.221 0.193 0.215 ...
## $ laseniorshalf : num [1:72864] 199 139 346 767 840 ...
## $ laseniorshalfshare : num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
## $ lawwhitehalf : num [1:72864] 1483 412 2115 3395 6299 ...
## $ lawwhitehalfshare : num [1:72864] 0.776 0.19 0.627 0.774 0.585 ...
## $ lablackhalf : num [1:72864] 184 945 528 170 1001 ...
## $ lablackhalfshare : num [1:72864] 0.0964 0.4356 0.1566 0.0388 0.0929 ...
## $ laasianhalf : num [1:72864] 12.72 4 9.03 14.87 208.98 ...
## $ laasianhalfshare : num [1:72864] 0.00665 0.00184 0.00268 0.00339 0.01941 ...
## $ lanhopihalf : num [1:72864] 0 0 1 2.64 5.2 ...
## $ lanhopihalfshare : num [1:72864] 0 0 0.000296 0.000602 0.000483 ...
## $ laaianhalf : num [1:72864] 13.97 4.72 9.96 8.03 37.99 ...
## $ laaianhalfshare : num [1:72864] 0.00731 0.00217 0.00295 0.00183 0.00353 ...
## $ laomultirhalf : num [1:72864] 38.3 44 101.7 60.4 226.8 ...
## $ laomultirhalfshare : num [1:72864] 0.0201 0.0203 0.0302 0.0138 0.0211 ...
## $ lahisphalf : num [1:72864] 39.7 34.6 76.4 61.4 276.9 ...
## $ lahisphalfshare : num [1:72864] 0.0208 0.0159 0.0227 0.014 0.0257 ...
## $ lahunvhalf : num [1:72864] 21.6 58.6 49.1 17.5 129.6 ...
## $ lahunvhalfshare : num [1:72864] 0.0311 0.0789 0.0391 0.0102 0.0317 ...
## $ lasnaphalf : num [1:72864] 101.9 127.4 100.2 67.7 339.1 ...
## $ lasnaphalfshare : num [1:72864] 0.147 0.1714 0.0798 0.0393 0.0831 ...
## $ lapop1 : num [1:72864] 1357 483 1418 1363 2643 ...
## $ lapop1share : num [1:72864] 0.71 0.223 0.42 0.311 0.246 ...
## $ lalowil : num [1:72864] 322 145 697 410 623 ...
## $ lalowilshare : num [1:72864] 0.1685 0.0669 0.2065 0.0934 0.0579 ...
## $ lakids1 : num [1:72864] 364 175 377 346 715 ...
## $ lakids1share : num [1:72864] 0.1902 0.0805 0.1118 0.0789 0.0664 ...
## $ laseniors1 : num [1:72864] 162 51 190 237 362 ...
## $ laseniors1share : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
## $ lawwhite1 : num [1:72864] 1162 128 1168 1233 2168 ...
## $ lawwhite1share : num [1:72864] 0.6076 0.0591 0.3463 0.2812 0.2014 ...
## $ lablack1 : num [1:72864] 147.5 335.4 202.5 80.9 343.2 ...
## $ lablack1share : num [1:72864] 0.0771 0.1546 0.06 0.0185 0.0319 ...
## $ laasian1 : num [1:72864] 11.02 1.59 7.82 6.88 47.49 ...
## $ laasian1share : num [1:72864] 0.005761 0.000733 0.002319 0.001567 0.004412 ...
## $ lanhopi1 : num [1:72864] 0 0 0 2 0.953 ...
## $ lanhopi1share : num [1:72864] 0.00 0.00 0.00 4.56e-04 8.85e-05 ...
## $ laaian1 : num [1:72864] 9.997 0.264 1.656 3.54 13.561 ...
## $ laaian1share : num [1:72864] 0.005228 0.000121 0.000491 0.000807 0.00126 ...
## $ laomultir1 : num [1:72864] 27.2 18 37.9 36.8 70.1 ...
## $ laomultir1share : num [1:72864] 0.01421 0.00828 0.01123 0.0084 0.00651 ...
## $ lahispl : num [1:72864] 29.7 11.2 31.2 29.9 85.8 ...
## $ lahisplshare : num [1:72864] 0.01554 0.00515 0.00926 0.00681 0.00797 ...
## $ lahunv1 : num [1:72864] 9.77 21.64 13.31 8.78 44.66 ...
## $ lahunv1share : num [1:72864] 0.0141 0.0291 0.0106 0.0051 0.0109 ...
## $ lasnap1 : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
## $ lasnap1share : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
## $ lapop10 : num [1:72864] 0 0 0 0 0 ...
## $ lapop10share : num [1:72864] 0 0 0 0 0 ...
## $ lalowil0 : num [1:72864] 0 0 0 0 0 ...
## $ lalowil0share : num [1:72864] 0 0 0 0 0 ...
## $ lakids10 : num [1:72864] 0 0 0 0 0 ...

```

```
## $ lakids10share      : num [1:72864] 0 0 0 0 0 ...
## $ laseniors10       : num [1:72864] 0 0 0 0 0 ...
## $ laseniors10share   : num [1:72864] 0 0 0 0 0 ...
## $ lawhite10         : num [1:72864] 0 0 0 0 0 ...
## $ lawhite10share     : num [1:72864] 0 0 0 0 0 ...
## $ lablack10         : num [1:72864] 0 0 0 0 0 ...
## $ lablack10share     : num [1:72864] 0 0 0 0 0 ...
## $ laasian10         : num [1:72864] 0 0 0 0 0 ...
## $ laasian10share     : num [1:72864] 0 0 0 0 0 ...
## $ lanhopi10         : num [1:72864] 0 0 0 0 0 ...
## $ lanhopi10share     : num [1:72864] 0 0 0 0 0 ...
## [list output truncated]
```

```
str(DATA2019)
```

```
## tibble [72,531 x 147] (S3: tbl_df/tbl/data.frame)
## $ CensusTract      : chr [1:72531] "01001020100" "01001020200" "01001020300" "01001020400" ...
## $ State            : chr [1:72531] "Alabama" "Alabama" "Alabama" "Alabama" ...
## $ County           : chr [1:72531] "Autauga County" "Autauga County" "Autauga County" "Autauga C
## $ Urban            : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ Pop2010          : num [1:72531] 1912 2170 3373 4386 10766 ...
## $ OHU2010          : num [1:72531] 693 743 1256 1722 4082 ...
## $ GroupQuartersFlag : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
## $ NUMGQTRS         : num [1:72531] 0 181 0 0 181 0 36 0 0 14 ...
## $ PCTGQTRS         : num [1:72531] 0 8.34 0 0 1.68 ...
## $ LILATracts_1And10 : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
## $ LILATracts_halfAnd10: num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
## $ LILATracts_1And20 : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
## $ LILATracts_Vehicle : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
## $ HUNVFlag         : num [1:72531] 0 0 0 0 1 0 0 0 1 0 ...
## $ LowIncomeTracts   : num [1:72531] 0 1 0 0 0 1 1 0 0 0 ...
## $ PovertyRate       : num [1:72531] 11.34 17.88 15.05 2.85 15.15 ...
## $ MedianFamilyIncome : num [1:72531] 81250 49000 62609 70607 96334 ...
## $ LA1and10          : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ LAhalfand10       : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ LA1and20          : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ LATracts_half     : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ LATracts1         : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ LATracts10        : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
## $ LATracts20        : num [1:72531] 0 0 0 0 0 0 0 0 0 0 ...
## $ LATractsVehicle_20 : num [1:72531] 0 0 0 0 1 0 0 0 1 0 ...
## $ LAPOP1_10         : num [1:72531] 1896 1261 1552 1363 2643 ...
## $ LAPOP05_10        : num [1:72531] 1912 2170 2857 3651 7778 ...
## $ LAPOP1_20         : num [1:72531] 1896 1261 1552 1363 2643 ...
## $ LALOWI1_10        : num [1:72531] 461 604 478 343 586 ...
## $ LALOWI05_10       : num [1:72531] 467 962 971 893 1719 ...
## $ LALOWI1_20        : num [1:72531] 461 604 478 343 586 ...
## $ lapophalf         : num [1:72531] 1912 2170 2857 3651 7778 ...
## $ lapophalfshare    : num [1:72531] 100 100 84.7 83.2 72.2 ...
## $ lalowihalf        : num [1:72531] 467 962 971 893 1719 ...
## $ lalowihalfshare    : num [1:72531] 24.4 44.3 28.8 20.4 16 ...
## $ lakidshalf        : num [1:72531] 507 606 771 847 2309 ...
## $ lakidshalfshare    : num [1:72531] 26.5 27.9 22.9 19.3 21.5 ...
## $ laseniorshalf     : num [1:72531] 221 214 358 767 840 ...
## $ laseniorshalfshare : num [1:72531] 11.56 9.86 10.6 17.48 7.8 ...
```

```

## $ lawwhitehalf      : num [1:72531] 1622 888 2177 3395 6299 ...
## $ lawwhitehalfshare  : num [1:72531] 84.8 40.9 64.5 77.4 58.5 ...
## $ lablackhalf       : num [1:72531] 217 1217 554 170 1001 ...
## $ lablackhalfshare   : num [1:72531] 11.35 56.08 16.43 3.88 9.29 ...
## $ laasianhalf       : num [1:72531] 14 5 10.3 14.9 209 ...
## $ laasianhalfshare   : num [1:72531] 0.732 0.23 0.304 0.339 1.941 ...
## $ lanhopihalf       : num [1:72531] 0 0 1 2.64 5.2 ...
## $ lanhopihalfshare   : num [1:72531] 0 0 0.0296 0.0602 0.0483 ...
## $ laaianhalf        : num [1:72531] 14 5 10.24 8.03 37.99 ...
## $ laaianhalfshare    : num [1:72531] 0.732 0.23 0.304 0.183 0.353 ...
## $ laomultirhalf     : num [1:72531] 45 55 104.5 60.4 226.8 ...
## $ laomultirhalfshare : num [1:72531] 2.35 2.53 3.1 1.38 2.11 ...
## $ lahisphalf        : num [1:72531] 44 75 77.6 61.4 276.9 ...
## $ lahisphalfshare    : num [1:72531] 2.3 3.46 2.3 1.4 2.57 ...
## $ lahunvhalf        : num [1:72531] 5.49 92.67 38.77 19.42 163.89 ...
## $ lahunvhalfshare    : num [1:72531] 0.792 12.473 3.087 1.128 4.015 ...
## $ lasnaphalf        : num [1:72531] 92.4 161.2 138.8 84.1 235 ...
## $ lasnaphalfshare    : num [1:72531] 13.33 21.7 11.05 4.88 5.76 ...
## $ lapop1            : num [1:72531] 1896 1261 1552 1363 2643 ...
## $ lapop1share       : num [1:72531] 99.2 58.1 46 31.1 24.6 ...
## $ lalowil           : num [1:72531] 461 604 478 343 586 ...
## $ lalowilshare      : num [1:72531] 24.11 27.83 14.18 7.83 5.45 ...
## $ lakids1           : num [1:72531] 504 406 416 346 715 ...
## $ lakids1share      : num [1:72531] 26.33 18.69 12.34 7.89 6.64 ...
## $ laseniors1        : num [1:72531] 219 127 201 237 362 ...
## $ laseniors1share    : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
## $ lawwhite1         : num [1:72531] 1611 357 1242 1233 2168 ...
## $ lawwhite1share     : num [1:72531] 84.3 16.4 36.8 28.1 20.1 ...
## $ lablack1          : num [1:72531] 213.6 854.2 255.1 80.9 343.2 ...
## $ lablack1share      : num [1:72531] 11.17 39.36 7.56 1.85 3.19 ...
## $ laasian1          : num [1:72531] 13.8 4 8.03 6.88 47.49 ...
## $ laasian1share      : num [1:72531] 0.722 0.184 0.238 0.157 0.441 ...
## $ lanhopi1         : num [1:72531] 0 0 0 2 0.953 ...
## $ lanhopi1share      : num [1:72531] 0 0 0 0.0456 0.00885 ...
## $ laaian1           : num [1:72531] 14 4.4 2.08 3.54 13.56 ...
## $ laaian1share       : num [1:72531] 0.7322 0.2026 0.0615 0.0807 0.126 ...
## $ laomultir1        : num [1:72531] 44.1 41.8 44.8 36.8 70.1 ...
## $ laomultir1share    : num [1:72531] 2.307 1.926 1.328 0.84 0.651 ...
## $ lahispl           : num [1:72531] 43.4 33 36.3 29.9 85.8 ...
## $ lahisplshare       : num [1:72531] 2.268 1.52 1.075 0.681 0.797 ...
## $ lahunv1           : num [1:72531] 5.49 66.9 0 7.94 55.21 ...
## $ lahunv1share       : num [1:72531] 0.792 9.005 0 0.461 1.353 ...
## $ lasnap1           : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
## $ lasnap1share       : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
## $ lapop10           : num [1:72531] NA NA NA NA NA ...
## $ lapop10share       : num [1:72531] NA NA NA NA NA ...
## $ lalowil0           : num [1:72531] NA NA NA NA NA ...
## $ lalowil0share      : num [1:72531] NA NA NA NA NA ...
## $ lakids10           : num [1:72531] NA NA NA NA NA ...
## $ lakids10share      : num [1:72531] NA NA NA NA NA ...
## $ laseniors10        : num [1:72531] NA NA NA NA NA ...
## $ laseniors10share   : num [1:72531] NA NA NA NA NA ...
## $ lawwhite10         : num [1:72531] NA NA NA NA NA ...
## $ lawwhite10share    : num [1:72531] NA NA NA NA NA ...

```



```
## $ lablack10          : num [1:72531] NA NA NA NA NA ...
## $ lablack10share     : num [1:72531] NA NA NA NA NA ...
## $ laasian10          : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
## $ laasian10share     : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
## $ lanhopi10          : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
## $ lanhopi10share     : num [1:72531] NA NA NA NA NA NA NA 0 0 0 ...
## [list output truncated]
```

```
# To Unify variable names
```

```
names(DATA2015) <- tolower(names(DATA2015))
names(DATA2019) <- tolower(names(DATA2019))
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
# Select variables we need
```

```
data2015_clean <- DATA2015 %>%
```

```
  select(
    censustract,
    state,
    county,
    urban,
    pop2010,
    ohu2010,
    laseniors1,
    laseniors10,
    lasnap1,
    laseniors1share,
    laseniors10share,
    lasnap1share,
    lasnap10share,
    lasnap10,
    laseniors20,
    laseniorshalf,
    laseniorshalfshare,
```

```

    tractseniors,
    tractsnap,
  )

data2019_clean <- DATA2019 %>%
  select(
    censustract,
    state,
    county,
    urban,
    pop2010,
    ohu2010,
    laseniors1,
    laseniors10,
    laseniors1share,
    laseniors10share,
    lasnap1share,
    lasnap10share,
    lasnap1,
    lasnap10,
    laseniors20,
    laseniorshalf,
    laseniorshalfshare,
    tractseniors,
    tractsnap,
  )

```

```

numeric_vars <- c(
  "urban",
  "pop2010",
  "ohu2010",
  "laseniors1",
  "laseniors10",
  "laseniors1share",
  "laseniors10share",
  "lasnap1",
  "lasnap10",
  "lasnap1share",
  "lasnap10share",
  "tractseniors",
  "tractsnap"
)

data2015_clean <- data2015_clean %>%
  mutate(
    across(
      any_of(numeric_vars),
      as.numeric
    )
  )

data2019_clean <- data2019_clean %>%
  mutate(
    across(

```

```

    any_of(numeric_vars),
    as.numeric
  )
)

combined_data <- bind_rows(
  data2015_clean,
  data2019_clean
)

combined_data

```

```

## # A tibble: 145,395 x 19
##   censustract state county urban pop2010 ohu2010 laseniors1 laseniors10 lasnap1
##   <chr>         <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 01001020100 Alab~ Autau~ 1 1912 693 162. 0 79.5
## 2 01001020200 Alab~ Autau~ 1 2170 743 51.0 0 41.7
## 3 01001020300 Alab~ Autau~ 1 3373 1256 190. 0 50.3
## 4 01001020400 Alab~ Autau~ 1 4386 1722 237. 0 24.4
## 5 01001020500 Alab~ Autau~ 1 10766 4082 362. 0 120.
## 6 01001020600 Alab~ Autau~ 1 3668 1311 264. 0 87.5
## 7 01001020700 Alab~ Autau~ 1 2891 1188 109. 0 110.
## 8 01001020801 Alab~ Autau~ 0 3081 1074 341. 0 105.
## 9 01001020802 Alab~ Autau~ 0 10435 3694 1060. 24.4 365.
## 10 01001020900 Alab~ Autau~ 0 5675 2067 630. 242. 391.
## # i 145,385 more rows
## # i 10 more variables: laseniors1share <dbl>, laseniors10share <dbl>,
## # lasnap1share <dbl>, lasnap10share <dbl>, lasnap10 <dbl>, laseniors20 <dbl>,
## # laseniorshalf <dbl>, laseniorshalfshare <dbl>, tractseniors <dbl>,
## # tractsnap <dbl>

```

```

sapply(
  data2015_clean[numeric_vars],
  class
)

```

```

##      urban      pop2010      ohu2010      laseniors1
##      "numeric"      "numeric"      "numeric"      "numeric"
##      laseniors10 laseniors1share laseniors10share      lasnap1
##      "numeric"      "numeric"      "numeric"      "numeric"
##      lasnap10      lasnap1share      lasnap10share      tractseniors
##      "numeric"      "numeric"      "numeric"      "numeric"
##      tractsnap
##      "numeric"

```

```

sapply(
  data2019_clean[numeric_vars],
  class
)

```

```

##      urban      pop2010      ohu2010      laseniors1
##      "numeric"      "numeric"      "numeric"      "numeric"
##      laseniors10 laseniors1share laseniors10share      lasnap1
##      "numeric"      "numeric"      "numeric"      "numeric"
##      lasnap10      lasnap1share      lasnap10share      tractseniors

```

```
##      "numeric"      "numeric"      "numeric"      "numeric"
##      tractsnap
##      "numeric"
```

```
data2015_clean <- data2015_clean %>%
  mutate(year = 2015)
```

```
data2019_clean <- data2019_clean %>%
  mutate(year = 2019)
```

```
#Create Urban/Rural
```

```
data2015_clean <- data2015_clean %>%
  mutate(
    year = 2015,
    area = if_else(
      urban == 1,
      "Urban",
      "Rural",
      missing = NA_character_
    )
  )
```

```
data2019_clean <- data2019_clean %>%
  mutate(
    year = 2019,
    area = if_else(
      urban == 1,
      "Urban",
      "Rural",
      missing = NA_character_
    )
  )
```

```
# 2015 of senior and snap
```

```
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_n = case_when(
      urban == 1 ~ laseniors1,
      urban == 0 ~ laseniors10,
      TRUE ~ NA_real_
    ),
    snap_la_n = case_when(
      urban == 1 ~ lasnap1,
      urban == 0 ~ lasnap10,
      TRUE ~ NA_real_
    )
  )
```

```
# 2019 of senior and snap
```

```
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_n = case_when(
      urban == 1 ~ laseniors1,
      urban == 0 ~ laseniors10,
```

```

    TRUE ~ NA_real_
  ),

  snap_la_n = case_when(
    urban == 1 ~ lasnap1,
    urban == 0 ~ lasnap10,
    TRUE ~ NA_real_
  )
)

# Show precentage
# 2015
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_pct = case_when(
      urban == 1 ~ laseniors1share * 100,
      urban == 0 ~ laseniors10share * 100,
      TRUE ~ NA_real_
    ),

    snap_la_pct = case_when(
      urban == 1 ~ lasnap1share * 100,
      urban == 0 ~ lasnap10share * 100,
      TRUE ~ NA_real_
    )
  )

# 2019
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_pct = case_when(
      urban == 1 ~ laseniors1share,
      urban == 0 ~ laseniors10share,
      TRUE ~ NA_real_
    ),

    snap_la_pct = case_when(
      urban == 1 ~ lasnap1share,
      urban == 0 ~ lasnap10share,
      TRUE ~ NA_real_
    )
  )

# Seniors
data2015_clean <- data2015_clean %>%
  mutate(
    senior_la_n = if_else(
      urban == 1,
      laseniors1,
      laseniors10
    )
  )

# SNAP

```

```

data2015_clean <- data2015_clean %>%
  mutate(
    snap_la_n = if_else(
      urban == 1,
      lasnap1,
      lasnap10
    )
  )

# Seniors
data2019_clean <- data2019_clean %>%
  mutate(
    senior_la_n = if_else(
      urban == 1,
      laseniors1,
      laseniors10
    )
  )

# SNAP
data2019_clean <- data2019_clean %>%
  mutate(
    snap_la_n = if_else(
      urban == 1,
      lasnap1,
      lasnap10
    )
  )

```

```
summary(data2015_clean)
```

```

##  censustract      state      county      urban
##  Length:72864    Length:72864    Length:72864    Min.   :0.0000
##  Class :character Class :character Class :character 1st Qu.:1.0000
##  Mode  :character Mode  :character Mode  :character Median :1.0000
##                                     Mean  :0.7572
##                                     3rd Qu.:1.0000
##                                     Max.   :1.0000
##
##  pop2010      ohu2010      laseniors1      laseniors10
##  Min.   :    0    Min.   :    0    Min.   :    0.0    Min.   :    0.0
##  1st Qu.: 2884    1st Qu.: 1102    1st Qu.:    0.0    1st Qu.:    0.0
##  Median : 4002    Median : 1521    Median :   111.2    Median :    0.0
##  Mean   : 4237    Mean   : 1602    Mean   :   230.6    Mean   :   12.1
##  3rd Qu.: 5323    3rd Qu.: 2018    3rd Qu.:   390.6    3rd Qu.:    0.0
##  Max.   :37452    Max.   :16043    Max.   :10225.7    Max.   :2531.0
##
##  lasnap1      laseniors1share  laseniors10share  lasnap1share
##  Min.   :    0.00    Min.   :0.000000    Min.   :0.000000    Min.   :0.000000
##  1st Qu.:    0.00    1st Qu.:0.000000    1st Qu.:0.000000    1st Qu.:0.000000
##  Median :   20.11    Median :0.02761    Median :0.000000    Median :0.01323
##  Mean   :   71.48    Mean   :0.05636    Mean   :0.00413    Mean   :0.04602
##  3rd Qu.:  103.09    3rd Qu.:0.09887    3rd Qu.:0.000000    3rd Qu.:0.06732
##  Max.   : 1521.24    Max.   :1.00000    Max.   :0.57680    Max.   :1.00000
##
##  lasnap10share  lasnap10      laseniors20      laseniorshalf
##  Min.   :0.000000    Min.   :    0.00    Min.   :    0.000    Min.   :    0.0

```

```
## 1st Qu.:0.000000 1st Qu.: 0.00 1st Qu.: 0.000 1st Qu.: 153.7
## Median :0.000000 Median : 0.00 Median : 0.000 Median : 346.0
## Mean :0.003116 Mean : 3.71 Mean : 1.492 Mean : 395.2
## 3rd Qu.:0.000000 3rd Qu.: 0.00 3rd Qu.: 0.000 3rd Qu.: 567.2
## Max. :0.888889 Max. :1108.69 Max. :2081.129 Max. :15260.1
## laseniorshalfshare tractseniors tractsnap year
## Min. :0.00000 Min. : 0.0 Min. : 0 Min. :2015
## 1st Qu.:0.04248 1st Qu.: 318.0 1st Qu.: 69 1st Qu.:2015
## Median :0.09145 Median : 495.0 Median : 158 Median :2015
## Mean :0.09642 Mean : 552.6 Mean : 207 Mean :2015
## 3rd Qu.:0.13747 3rd Qu.: 716.0 3rd Qu.: 293 3rd Qu.:2015
## Max. :1.00000 Max. :17271.0 Max. :2152 Max. :2015
## area senior_la_n snap_la_n senior_la_pct
## Length:72864 Min. : 0.000 Min. : 0.0000 Min. : 0.0000
## Class :character 1st Qu.: 0.000 1st Qu.: 0.0000 1st Qu.: 0.0000
## Mode :character Median : 5.792 Median : 0.9336 Median : 0.1582
## Mean : 123.775 Mean : 38.2526 Mean : 2.9614
## 3rd Qu.: 167.168 3rd Qu.: 37.7681 3rd Qu.: 4.1252
## Max. :10225.695 Max. :1521.2415 Max. :100.0000
## snap_la_pct
## Min. : 0.00000
## 1st Qu.: 0.00000
## Median : 0.06551
## Mean : 2.48353
## 3rd Qu.: 2.41834
## Max. :100.00000
```

```
summary(data2019_clean)
```

```
## censustract state county urban
## Length:72531 Length:72531 Length:72531 Min. :0.0000
## Class :character Class :character Class :character 1st Qu.:1.0000
## Mode :character Mode :character Mode :character Median :1.0000
## Mean :0.7606
## 3rd Qu.:1.0000
## Max. :1.0000
##
## pop2010 ohu2010 laseniors1 laseniors10
## Min. : 1 Min. : 0 Min. : 0.00 Min. : 0.00
## 1st Qu.: 2899 1st Qu.: 1108 1st Qu.: 73.79 1st Qu.: 7.58
## Median : 4011 Median : 1525 Median : 259.28 Median : 51.70
## Mean : 4257 Mean : 1609 Mean : 319.30 Mean : 111.16
## 3rd Qu.: 5330 3rd Qu.: 2021 3rd Qu.: 484.00 3rd Qu.: 162.38
## Max. :37452 Max. :16043 Max. :10348.58 Max. :2531.00
## NA's :19989 NA's :64765
## laseniors1share laseniors10share lasnap1share lasnap10share
## Min. : 0.000 Min. : 0.00 Min. : 0.000 Min. : 0.00
## 1st Qu.: 1.879 1st Qu.: 0.20 1st Qu.: 0.794 1st Qu.: 0.11
## Median : 6.549 Median : 1.50 Median : 3.431 Median : 0.85
## Mean : 7.826 Mean : 3.81 Mean : 5.996 Mean : 2.67
## 3rd Qu.: 12.227 3rd Qu.: 5.30 3rd Qu.: 8.660 3rd Qu.: 3.01
## Max. :100.000 Max. :57.68 Max. :100.000 Max. :84.53
## NA's :19989 NA's :64765 NA's :19966 NA's :64666
## lasnap1 lasnap10 laseniors20 laseniorshalf
## Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.0
```

```
## 1st Qu.: 11.89 1st Qu.: 1.77 1st Qu.: 0.42 1st Qu.: 195.0
## Median : 52.93 Median : 11.33 Median : 12.40 Median : 372.7
## Mean : 92.52 Mean : 31.93 Mean : 63.29 Mean : 422.9
## 3rd Qu.: 133.53 3rd Qu.: 37.97 3rd Qu.: 72.97 3rd Qu.: 585.8
## Max. :1582.31 Max. :995.18 Max. :2081.13 Max. :15261.1
## NA's :19989 NA's :64765 NA's :71025 NA's :4568
## laseniorshalfshare tractseniors tractsnap year
## Min. : 0.000 Min. : 0.0 Min. : 0.0 Min. :2019
## 1st Qu.: 5.286 1st Qu.: 320.0 1st Qu.: 67.0 1st Qu.:2019
## Median : 9.768 Median : 497.0 Median : 152.0 Median :2019
## Mean : 10.326 Mean : 555.2 Mean : 201.8 Mean :2019
## 3rd Qu.: 14.080 3rd Qu.: 718.0 3rd Qu.: 282.0 3rd Qu.:2019
## Max. :100.000 Max. :17271.0 Max. :2175.0 Max. :2019
## NA's :4568 NA's :4 NA's :4
## area senior_la_n snap_la_n senior_la_pct
## Length:72531 Min. : 0.0 Min. : 0.000 Min. : 0.000
## Class :character 1st Qu.: 25.8 1st Qu.: 4.591 1st Qu.: 0.668
## Mode :character Median : 126.0 Median : 24.812 Median : 3.157
## Mean : 211.4 Mean : 61.941 Mean : 5.086
## 3rd Qu.: 308.0 3rd Qu.: 78.624 3rd Qu.: 7.530
## Max. :10348.6 Max. :1582.311 Max. :100.000
## NA's :29953 NA's :29953 NA's :29953
## snap_la_pct
## Min. : 0.000
## 1st Qu.: 0.304
## Median : 1.600
## Mean : 4.062
## 3rd Qu.: 4.869
## Max. :85.091
## NA's :29918
```

```
#data2019_clean <- data2019_clean %>%
# mutate(
#   across(
#     c(laseniors1, laseniors10, lasnap1, lasnap10,
#       laseniors20, laseniorshalf, laseniorshalfshare, senior_la_n, snap_la_n),
#     ~ as.numeric(ifelse(.x == "NULL", "0", .x))
#   )
# )
```

```
summary(data2019_clean)
```

```
## censustract state county urban
## Length:72531 Length:72531 Length:72531 Min. :0.0000
## Class :character Class :character Class :character 1st Qu.:1.0000
## Mode :character Mode :character Mode :character Median :1.0000
## Mean :0.7606
## 3rd Qu.:1.0000
## Max. :1.0000
##
## pop2010 ohu2010 laseniors1 laseniors10
## Min. : 1 Min. : 0 Min. : 0.00 Min. : 0.00
## 1st Qu.: 2899 1st Qu.: 1108 1st Qu.: 73.79 1st Qu.: 7.58
## Median : 4011 Median : 1525 Median : 259.28 Median : 51.70
## Mean : 4257 Mean : 1609 Mean : 319.30 Mean : 111.16
```



```

## 3rd Qu.: 5330 3rd Qu.: 2021 3rd Qu.: 484.00 3rd Qu.: 162.38
## Max. :37452 Max. :16043 Max. :10348.58 Max. :2531.00
## NA's :19989 NA's :64765
## laseniors1share laseniors10share lasnap1share lasnap10share
## Min. : 0.000 Min. : 0.00 Min. : 0.000 Min. : 0.00
## 1st Qu.: 1.879 1st Qu.: 0.20 1st Qu.: 0.794 1st Qu.: 0.11
## Median : 6.549 Median : 1.50 Median : 3.431 Median : 0.85
## Mean : 7.826 Mean : 3.81 Mean : 5.996 Mean : 2.67
## 3rd Qu.: 12.227 3rd Qu.: 5.30 3rd Qu.: 8.660 3rd Qu.: 3.01
## Max. :100.000 Max. :57.68 Max. :100.000 Max. :84.53
## NA's :19989 NA's :64765 NA's :19966 NA's :64666
## lasnap1 lasnap10 laseniors20 laseniorshalf
## Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.0
## 1st Qu.: 11.89 1st Qu.: 1.77 1st Qu.: 0.42 1st Qu.: 195.0
## Median : 52.93 Median : 11.33 Median : 12.40 Median : 372.7
## Mean : 92.52 Mean : 31.93 Mean : 63.29 Mean : 422.9
## 3rd Qu.: 133.53 3rd Qu.: 37.97 3rd Qu.: 72.97 3rd Qu.: 585.8
## Max. :1582.31 Max. :995.18 Max. :2081.13 Max. :15261.1
## NA's :19989 NA's :64765 NA's :71025 NA's :4568
## laseniorshalfshare tractseniors tractsnap year
## Min. : 0.000 Min. : 0.0 Min. : 0.0 Min. :2019
## 1st Qu.: 5.286 1st Qu.: 320.0 1st Qu.: 67.0 1st Qu.:2019
## Median : 9.768 Median : 497.0 Median : 152.0 Median :2019
## Mean : 10.326 Mean : 555.2 Mean : 201.8 Mean :2019
## 3rd Qu.: 14.080 3rd Qu.: 718.0 3rd Qu.: 282.0 3rd Qu.:2019
## Max. :100.000 Max. :17271.0 Max. :2175.0 Max. :2019
## NA's :4568 NA's :4 NA's :4
## area senior_la_n snap_la_n senior_la_pct
## Length:72531 Min. : 0.0 Min. : 0.000 Min. : 0.000
## Class :character 1st Qu.: 25.8 1st Qu.: 4.591 1st Qu.: 0.668
## Mode :character Median : 126.0 Median : 24.812 Median : 3.157
## Mean : 211.4 Mean : 61.941 Mean : 5.086
## 3rd Qu.: 308.0 3rd Qu.: 78.624 3rd Qu.: 7.530
## Max. :10348.6 Max. :1582.311 Max. :100.000
## NA's :29953 NA's :29953 NA's :29953
## snap_la_pct
## Min. : 0.000
## 1st Qu.: 0.304
## Median : 1.600
## Mean : 4.062
## 3rd Qu.: 4.869
## Max. :85.091
## NA's :29918

```

```
library(knitr)
```

```

dataset_table <- data.frame(
  Dataset = c("2015 Food Access Research Atlas",
    "2019 Food Access Research Atlas"),
  Observations = c(72864, 72531),
  Variables = c(147, 147),
  Unit_of_Analysis = c("Census tract", "Census tract"),
  Main_Data_Types = c(
    "Geographic identifiers, urban/rural indicator, population counts, food-access measures",

```

```

    "Geographic identifiers, urban/rural indicator, population counts, food-access measures"
  )
)

kable(
  dataset_table,
  caption = "Structure of the 2015 and 2019 Food Access Research Atlas datasets",
  col.names = c(
    "Dataset",
    "Observations",
    "Variables",
    "Unit of Analysis",
    "Main Data Types"
  ),
  align = c("l", "r", "r", "l", "l")
)

```

Table 1: Structure of the 2015 and 2019 Food Access Research Atlas datasets

Dataset	Observations	Variables	Unit of Analysis	Main Data Types
2015 Food Access Research Atlas	72864	147	Census tract	Geographic identifiers, urban/rural indicator, population counts, food-access measures
2019 Food Access Research Atlas	72531	147	Census tract	Geographic identifiers, urban/rural indicator, population counts, food-access measures

```

summary_stats <- bind_rows(

  data2015_clean %>%
    transmute(
      year = 2015,
      Senior = senior_la_pct,
      SNAP = snap_la_pct
    ),

  data2019_clean %>%
    transmute(
      year = 2019,
      Senior = senior_la_pct,
      SNAP = snap_la_pct
    )

) %>%

pivot_longer(
  cols = c(Senior, SNAP),
  names_to = "Group",
  values_to = "Low_Access_Pct"
) %>%

group_by(year, Group) %>%

```

```

summarise(
  Min = min(Low_Access_Pct, na.rm = TRUE),
  Q1 = quantile(Low_Access_Pct, 0.25, na.rm = TRUE),
  Median = median(Low_Access_Pct, na.rm = TRUE),
  Mean = mean(Low_Access_Pct, na.rm = TRUE),
  Q3 = quantile(Low_Access_Pct, 0.75, na.rm = TRUE),
  Max = max(Low_Access_Pct, na.rm = TRUE),
  Missing = sum(is.na(Low_Access_Pct)),
  .groups = "drop"
) %>%

mutate(
  across(
    c(Min, Q1, Median, Mean, Q3, Max),
    ~ round(.x, 2)
  )
)

summary_stats

## # A tibble: 4 x 9
##   year Group   Min    Q1 Median  Mean    Q3   Max Missing
##   <dbl> <chr>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <int>
## 1  2015 SNAP      0  0      0.07  2.48  2.42 100      0
## 2  2015 Senior    0  0      0.16  2.96  4.13 100      0
## 3  2019 SNAP      0  0.3    1.6   4.06  4.87 85.1 29918
## 4  2019 Senior    0  0.67   3.16  5.09  7.53 100 29953

kable(
  summary_stats,
  caption = "Summary Statistics for Senior and SNAP Low-Access Shares, 2015 and 2019",
  col.names = c(
    "Year",
    "Group",
    "Min (%)",
    "Q1 (%)",
    "Median (%)",
    "Mean (%)",
    "Q3 (%)",
    "Max (%)",
    "Missing"
  ),
  align = c(
    "c", "l",
    "r", "r", "r", "r", "r", "r", "r"
  )
)

```

Table 2: Summary Statistics for Senior and SNAP Low-Access Shares, 2015 and 2019

Year	Group	Min (%)	Q1 (%)	Median (%)	Mean (%)	Q3 (%)	Max (%)	Missing
2015	SNAP	0	0.00	0.07	2.48	2.42	100.00	0
2015	Senior	0	0.00	0.16	2.96	4.13	100.00	0

Year	Group	Min (%)	Q1 (%)	Median (%)	Mean (%)	Q3 (%)	Max (%)	Missing
2019	SNAP	0	0.30	1.60	4.06	4.87	85.09	29918
2019	Senior	0	0.67	3.16	5.09	7.53	100.00	29953

```
# Check whether over 100
```

```
data2015_clean %>%
```

```
  summarise(
    senior_over_100 = sum(senior_la_pct > 100, na.rm = TRUE),
    snap_over_100 = sum(snap_la_pct > 100, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 2
```

```
##   senior_over_100 snap_over_100
```

```
##       <int>         <int>
```

```
## 1           0           1
```

```
data2019_clean %>%
```

```
  summarise(
    senior_over_100 = sum(senior_la_pct > 100, na.rm = TRUE),
    snap_over_100 = sum(snap_la_pct > 100, na.rm = TRUE)
  )
```

```
## # A tibble: 1 x 2
```

```
##   senior_over_100 snap_over_100
```

```
##       <int>         <int>
```

```
## 1           0           0
```

```
data2015_clean %>%
```

```
  filter(snap_la_pct > 100) %>%
```

```
  select(
```

```
    censustract,
```

```
    urban,
```

```
    area,
```

```
    lasnap1share,
```

```
    lasnap10share,
```

```
    snap_la_pct
```

```
  )
```

```
## # A tibble: 1 x 6
```

```
##   censustract urban area  lasnap1share lasnap10share snap_la_pct
```

```
##   <chr>      <dbl> <chr>      <dbl>      <dbl>      <dbl>
```

```
## 1 22079980000 1 Urban      1.00        0        100.
```

```
data2019_clean %>%
```

```
  filter(snap_la_pct > 100) %>%
```

```
  select(
```

```
    censustract,
```

```
    urban,
```

```
    area,
```

```
    lasnap1share,
```

```
    lasnap10share,
```

```
    snap_la_pct
```

```
  )
```

```
## # A tibble: 0 x 6
```

```
## # i 6 variables: censustract <chr>, urban <dbl>, area <chr>,
```

```
## # lasnap1share <dbl>, lasnap10share <dbl>, snap_la_pct <dbl>
```

```
options(digits = 15)
```

```
data2015_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 1 x 6
```

```
##   censustract urban area lasnap1share lasnap10share snap_la_pct
##   <chr>         <dbl> <chr>         <dbl>         <dbl>         <dbl>
## 1 22079980000     1 Urban           1.00           0           100.
```

```
data2019_clean %>%
  filter(snap_la_pct > 100) %>%
  select(
    censustract,
    urban,
    area,
    lasnap1share,
    lasnap10share,
    snap_la_pct
  )
```

```
## # A tibble: 0 x 6
```

```
## # i 6 variables: censustract <chr>, urban <dbl>, area <chr>,
## # lasnap1share <dbl>, lasnap10share <dbl>, snap_la_pct <dbl>
```

```
# Show precentage
#data2015_clean <- data2015_clean %>%
# mutate(
#   senior_la_pct = senior_la_rate * 100,
#   snap_la_pct = snap_la_rate * 100
# )
```

```
#data2019_clean <- data2019_clean %>%
# mutate(
#   senior_la_pct = senior_la_rate * 100,
#   snap_la_pct = snap_la_rate * 100
# )
```

```
combined_data <- bind_rows(
  data2015_clean,
  data2019_clean
)
combined_data
```

```
## # A tibble: 145,395 x 25
```

```
##   censustract state county urban pop2010 ohu2010 laseniors1 laseniors10 lasnap1
##   <chr>         <chr> <chr>  <dbl>  <dbl>  <dbl>      <dbl>      <dbl>  <dbl>
```

```
## 1 01001020100 Alab~ Autau~ 1 1912 693 162. 0 79.5
## 2 01001020200 Alab~ Autau~ 1 2170 743 51.0 0 41.7
## 3 01001020300 Alab~ Autau~ 1 3373 1256 190. 0 50.3
## 4 01001020400 Alab~ Autau~ 1 4386 1722 237. 0 24.4
## 5 01001020500 Alab~ Autau~ 1 10766 4082 362. 0 120.
## 6 01001020600 Alab~ Autau~ 1 3668 1311 264. 0 87.5
## 7 01001020700 Alab~ Autau~ 1 2891 1188 109. 0 110.
## 8 01001020801 Alab~ Autau~ 0 3081 1074 341. 0 105.
## 9 01001020802 Alab~ Autau~ 0 10435 3694 1060. 24.4 365.
## 10 01001020900 Alab~ Autau~ 0 5675 2067 630. 242. 391.
## # i 145,385 more rows
## # i 16 more variables: laseniors1share <dbl>, laseniors10share <dbl>,
## # lasnap1share <dbl>, lasnap10share <dbl>, lasnap10 <dbl>, laseniors20 <dbl>,
## # laseniorshalf <dbl>, laseniorshalfshare <dbl>, tractseniors <dbl>,
## # tractsnap <dbl>, year <dbl>, area <chr>, senior_la_n <dbl>,
## # snap_la_n <dbl>, senior_la_pct <dbl>, snap_la_pct <dbl>
```

```
dim(combined_data)
```

```
## [1] 145395 25
```

```
table(combined_data$year)
```

```
##
## 2015 2019
## 72864 72531
```

```
table(combined_data$area, useNA = "ifany")
```

```
##
## Rural Urban
## 35054 110341
```

Explain: After harmonisation, the 2015 and 2019 datasets were combined into a dataset containing 145,395 census-tract observations. Of these, 110,341 were classified as urban and 35,054 as rural, with no missing values in the urban–rural classification.

```
summary_table <- combined_data %>%
  group_by(year, area) %>%
  summarise(
    n_tracts = n(),

    n_valid_senior =
      sum(!is.na(senior_la_pct)),

    n_valid_snap =
      sum(!is.na(snap_la_pct)),

    mean_senior_pct =
      mean(senior_la_pct, na.rm = TRUE),

    median_senior_pct =
      median(senior_la_pct, na.rm = TRUE),

    mean_snap_pct =
      mean(snap_la_pct, na.rm = TRUE),
```

```

median_snap_pct =
  median(snap_la_pct, na.rm = TRUE),

  .groups = "drop"
)
#summary_table <- combined_data %>%
# group_by(year, area) %>%
# summarise(n_tracts = n(),
#   senior_la_rate =
#     sum(senior_la_n, na.rm = TRUE) /
#     sum(tractseniors, na.rm = TRUE),
#   snap_la_rate =
#     sum(snap_la_n, na.rm = TRUE) /
#     sum(tractsnap, na.rm = TRUE),
#   .groups = "drop"
# ) %>%

# mutate(
#   senior_la_pct =
#     senior_la_rate * 100,
#   snap_la_pct =
#     snap_la_rate * 100
# )

summary_table

## # A tibble: 4 x 9
##   year area  n_tracts n_valid_senior n_valid_snap mean_senior_pct
##   <dbl> <chr>   <int>         <int>         <int>         <dbl>
## 1  2015 Rural   17692         17692         17692         1.68
## 2  2015 Urban   55172         55172         55172         3.37
## 3  2019 Rural   17362          7392          7404         3.96
## 4  2019 Urban   55169         35186         35209         5.32
## # i 3 more variables: median_senior_pct <dbl>, mean_snap_pct <dbl>,
## #   median_snap_pct <dbl>

summary_table_report <- summary_table %>%
  mutate(
    across(
      c(
        mean_senior_pct,
        median_senior_pct,
        mean_snap_pct,
        median_snap_pct
      ),
      ~ round(.x, 2)
    )
  )

summary_table_report

## # A tibble: 4 x 9
##   year area  n_tracts n_valid_senior n_valid_snap mean_senior_pct
##   <dbl> <chr>   <int>         <int>         <int>         <dbl>

```

```
## 1 2015 Rural 17692 17692 17692 1.68
## 2 2015 Urban 55172 55172 55172 3.37
## 3 2019 Rural 17362 7392 7404 3.96
## 4 2019 Urban 55169 35186 35209 5.32
## # i 3 more variables: median_senior_pct <dbl>, mean_snap_pct <dbl>,
## # median_snap_pct <dbl>
```

```
dim(data2015_clean)
```

```
## [1] 72864 25
```

```
names(data2015_clean)
```

```
## [1] "censustract" "state" "county"
## [4] "urban" "pop2010" "ohu2010"
## [7] "laseniors1" "laseniors10" "lasnap1"
## [10] "laseniors1share" "laseniors10share" "lasnap1share"
## [13] "lasnap10share" "lasnap10" "laseniors20"
## [16] "laseniorshalf" "laseniorshalfshare" "tractseniors"
## [19] "tractsnap" "year" "area"
## [22] "senior_la_n" "snap_la_n" "senior_la_pct"
## [25] "snap_la_pct"
```

```
str(data2015_clean)
```

```
## tibble [72,864 x 25] (S3: tbl_df/tbl/data.frame)
## $ censustract : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
## $ state : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
## $ county : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
## $ urban : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
## $ pop2010 : num [1:72864] 1912 2170 3373 4386 10766 ...
## $ ohu2010 : num [1:72864] 693 743 1256 1722 4082 ...
## $ laseniors1 : num [1:72864] 162 51 190 237 362 ...
## $ laseniors10 : num [1:72864] 0 0 0 0 0 ...
## $ lasnap1 : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
## $ laseniors1share : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
## $ laseniors10share : num [1:72864] 0 0 0 0 0 ...
## $ lasnap1share : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
## $ lasnap10share : num [1:72864] 0 0 0 0 0 ...
## $ lasnap10 : num [1:72864] 0 0 0 0 0 ...
## $ laseniors20 : num [1:72864] 0 0 0 0 0 0 0 0 0 ...
## $ laseniorshalf : num [1:72864] 199 139 346 767 840 ...
## $ laseniorshalfshare : num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
## $ tractseniors : num [1:72864] 221 214 439 904 1126 ...
## $ tractsnap : num [1:72864] 112 202 120 82 488 118 218 115 367 394 ...
## $ year : num [1:72864] 2015 2015 2015 2015 2015 ...
## $ area : chr [1:72864] "Urban" "Urban" "Urban" "Urban" ...
## $ senior_la_n : num [1:72864] 162 51 190 237 362 ...
## $ snap_la_n : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
## $ senior_la_pct : num [1:72864] 8.5 2.35 5.63 5.39 3.36 ...
## $ snap_la_pct : num [1:72864] 11.48 5.61 4 1.42 2.94 ...
```

```
summary(data2015_clean)
```

```
## censustract state county
## Length:72864 Length:72864 Length:72864
## Class :character Class :character Class :character
```



```

## Mode :character Mode :character Mode :character
##
##
##
##      urban      pop2010      ohu2010
## Min. :0.000000000000 Min. : 0.0000000 Min. : 0.00000000
## 1st Qu.:1.000000000000 1st Qu.: 2883.7500000 1st Qu.: 1102.00000000
## Median :1.000000000000 Median : 4002.0000000 Median : 1521.00000000
## Mean :0.757191480018 Mean : 4237.2850516 Mean : 1601.83756039
## 3rd Qu.:1.000000000000 3rd Qu.: 5323.0000000 3rd Qu.: 2018.00000000
## Max. :1.000000000000 Max. :37452.0000000 Max. :16043.00000000
## laseniors1 laseniors10 lasnap1
## Min. : 0.000000000 Min. : 0.0000000000 Min. : 0.0000000000
## 1st Qu.: 0.000000000 1st Qu.: 0.0000000000 1st Qu.: 0.0000000000
## Median : 111.152876843 Median : 0.0000000000 Median : 20.1084352526
## Mean : 230.558283934 Mean : 12.0994974777 Mean : 71.4784359749
## 3rd Qu.: 390.563446517 3rd Qu.: 0.0000000000 3rd Qu.: 103.0891161450
## Max. :10225.694634300 Max. :2530.9999906900 Max. :1521.2415445500
## laseniors1share laseniors10share lasnap1share
## Min. :0.000000000000 Min. :0.000000000000000 Min. :0.0000000000000
## 1st Qu.:0.000000000000 1st Qu.:0.000000000000000 1st Qu.:0.0000000000000
## Median :0.0276110682663 Median :0.000000000000000 Median :0.0132343918389
## Mean :0.0563605601272 Mean :0.00413024682724 Mean :0.0460208855680
## 3rd Qu.:0.0988746660144 3rd Qu.:0.000000000000000 3rd Qu.:0.0673183309857
## Max. :1.0000000298000 Max. :0.57680036250900 Max. :1.0000000217700
## lasnap10share lasnap10
## Min. :0.00000000000000 Min. : 0.000000000000
## 1st Qu.:0.00000000000000 1st Qu.: 0.000000000000
## Median :0.00000000000000 Median : 0.000000000000
## Mean :0.00311638858043 Mean : 3.71003338017
## 3rd Qu.:0.00000000000000 3rd Qu.: 0.000000000000
## Max. :0.88888886780400 Max. :1108.68866587000
## laseniors20 laseniorshalf laseniorshalfshare
## Min. : 0.000000000000 Min. : 0.000000000 Min. :0.000000000000
## 1st Qu.: 0.000000000000 1st Qu.: 153.679179371 1st Qu.:0.0424771214540
## Median : 0.000000000000 Median : 345.999997497 Median :0.0914459654018
## Mean : 1.49147507485 Mean : 395.213484884 Mean :0.0964164394142
## 3rd Qu.: 0.000000000000 3rd Qu.: 567.160194513 3rd Qu.:0.1374729178860
## Max. :2081.12934891000 Max. :15260.141464200 Max. :1.0000000298000
## tractseniors tractsnap year
## Min. : 0.000000000 Min. : 0.000000000 Min. :2015
## 1st Qu.: 318.000000000 1st Qu.: 69.000000000 1st Qu.:2015
## Median : 495.000000000 Median : 158.000000000 Median :2015
## Mean : 552.645805885 Mean : 207.022205753 Mean :2015
## 3rd Qu.: 716.000000000 3rd Qu.: 293.000000000 3rd Qu.:2015
## Max. :17271.000000000 Max. :2152.000000000 Max. :2015
## area senior_la_n snap_la_n
## Length:72864 Min. : 0.000000000000 Min. : 0.000000000000
## Class :character 1st Qu.: 0.000000000000 1st Qu.: 0.000000000000
## Mode :character Median : 5.79194930115 Median : 0.933608777821
## Mean : 123.77540303500 Mean : 38.252579180900
## 3rd Qu.: 167.16781171100 3rd Qu.: 37.768104004800
## Max. :10225.69463430000 Max. :1521.241544550000
## senior_la_pct snap_la_pct

```

```
## Min. : 0.000000000000 Min. : 0.000000000000
## 1st Qu.: 0.000000000000 1st Qu.: 0.000000000000
## Median : 0.158221080176 Median : 0.0655059568322
## Mean : 2.961367271190 Mean : 2.4835267010000
## 3rd Qu.: 4.125176027870 3rd Qu.: 2.4183350131700
## Max. :100.000000000000 Max. :100.0000021770000
```

Data Cleaning and Missing Data.

```
dim(data2019_clean)
```

```
## [1] 72531 25
```

```
names(data2019_clean)
```

```
## [1] "censustract" "state" "county"
## [4] "urban" "pop2010" "ohu2010"
## [7] "laseniors1" "laseniors10" "laseniors1share"
## [10] "laseniors10share" "lasnap1share" "lasnap10share"
## [13] "lasnap1" "lasnap10" "laseniors20"
## [16] "laseniorshalf" "laseniorshalfshare" "tractseniors"
## [19] "tractsnap" "year" "area"
## [22] "senior_la_n" "snap_la_n" "senior_la_pct"
## [25] "snap_la_pct"
```

```
str(data2019_clean)
```

```
## tibble [72,531 x 25] (S3: tbl_df/tbl/data.frame)
## $ censustract : chr [1:72531] "01001020100" "01001020200" "01001020300" "01001020400" ...
## $ state : chr [1:72531] "Alabama" "Alabama" "Alabama" "Alabama" ...
## $ county : chr [1:72531] "Autauga County" "Autauga County" "Autauga County" "Autauga County" ...
## $ urban : num [1:72531] 1 1 1 1 1 1 1 0 0 0 ...
## $ pop2010 : num [1:72531] 1912 2170 3373 4386 10766 ...
## $ ohu2010 : num [1:72531] 693 743 1256 1722 4082 ...
## $ laseniors1 : num [1:72531] 219 127 201 237 362 ...
## $ laseniors10 : num [1:72531] NA NA NA NA NA ...
## $ laseniors1share : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
## $ laseniors10share : num [1:72531] NA NA NA NA NA ...
## $ lasnap1share : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
## $ lasnap10share : num [1:72531] NA NA NA NA NA ...
## $ lasnap1 : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
## $ lasnap10 : num [1:72531] NA NA NA NA NA ...
## $ laseniors20 : num [1:72531] NA NA NA NA NA NA NA NA NA NA ...
## $ laseniorshalf : num [1:72531] 221 214 358 767 840 ...
## $ laseniorshalfshare : num [1:72531] 11.56 9.86 10.6 17.48 7.8 ...
## $ tractseniors : num [1:72531] 221 214 439 904 1126 ...
## $ tractsnap : num [1:72531] 102 156 172 98 339 224 390 143 352 340 ...
## $ year : num [1:72531] 2019 2019 2019 2019 2019 ...
## $ area : chr [1:72531] "Urban" "Urban" "Urban" "Urban" ...
## $ senior_la_n : num [1:72531] 219 127 201 237 362 ...
## $ snap_la_n : num [1:72531] 91.6 96.2 73.7 30.3 83.1 ...
## $ senior_la_pct : num [1:72531] 11.44 5.83 5.96 5.39 3.36 ...
## $ snap_la_pct : num [1:72531] 13.22 12.95 5.87 1.76 2.04 ...
```

```
summary(data2019_clean)
```

```
## censustract state county
```

```

## Length:72531      Length:72531      Length:72531
## Class :character   Class :character   Class :character
## Mode :character    Mode :character    Mode :character
##
##
##
##
##      urban              pop2010              ohu2010
## Min.   :0.000000000000    Min.   : 1.00000000    Min.   : 0.00000000
## 1st Qu.:1.000000000000    1st Qu.: 2899.00000000    1st Qu.: 1108.00000000
## Median :1.000000000000    Median : 4011.00000000    Median : 1525.00000000
## Mean   :0.760626490742    Mean   : 4256.73902194    Mean   : 1609.19182143
## 3rd Qu.:1.000000000000    3rd Qu.: 5330.50000000    3rd Qu.: 2021.00000000
## Max.   :1.000000000000    Max.   :37452.00000000    Max.   :16043.00000000
##
##      laseniors1          laseniors10          laseniors1share
## Min.   : 0.000000000000    Min.   : 0.000000000000    Min.   : 0.000000000000
## 1st Qu.: 73.7881634476     1st Qu.: 7.5835490330     1st Qu.: 1.87904652929
## Median : 259.2818331850     Median : 51.6980597595     Median : 6.54900459429
## Mean   : 319.2990643370     Mean   : 111.1584019070     Mean   : 7.82625335695
## 3rd Qu.: 484.0000007010     3rd Qu.: 162.3787189970     3rd Qu.: 12.22684722590
## Max.   :10348.5798401000     Max.   :2530.9999906900     Max.   :100.00000298000
## NA's   :19989              NA's   :64765              NA's   :19989
##      laseniors10share    lasnap1share    lasnap10share
## Min.   : 0.000000000000    Min.   : 0.000000000000    Min.   : 0.000000000000
## 1st Qu.: 0.1969509957     1st Qu.: 0.79378705758     1st Qu.: 0.1106945645
## Median : 1.5001749784     Median : 3.43137257330     Median : 0.8474116327
## Mean   : 3.8071163129     Mean   : 5.99596583627     Mean   : 2.6684463901
## 3rd Qu.: 5.3015079945     3rd Qu.: 8.65955330459     3rd Qu.: 3.0099299123
## Max.   :57.6800362509     Max.   :100.00000199900     Max.   :84.5265517912
## NA's   :64765              NA's   :19966              NA's   :64666
##      lasnap1            lasnap10            laseniors20
## Min.   : 0.000000000000    Min.   : 0.000000000000    Min.   : 0.000000000000
## 1st Qu.: 11.8921063110     1st Qu.: 1.7691137766     1st Qu.: 0.4210301405
## Median : 52.9257316770     Median : 11.3287673211     Median : 12.4007253413
## Mean   : 92.5222140142     Mean   : 31.9297830959     Mean   : 63.2935971552
## 3rd Qu.: 133.5308377880     3rd Qu.: 37.9660582705     3rd Qu.: 72.9691026808
## Max.   :1582.3107130700     Max.   :995.1762471410     Max.   :2081.1293489100
## NA's   :19989              NA's   :64765              NA's   :71025
##      laseniorshalf      laseniorshalfshare    tractseniors
## Min.   : 0.000000000000    Min.   : 0.000000000000    Min.   : 0.000000000000
## 1st Qu.: 195.009266620     1st Qu.: 5.28589415698     1st Qu.: 320.0000000000
## Median : 372.666346259     Median : 9.76757168611     Median : 497.0000000000
## Mean   : 422.927560834     Mean   : 10.32568201080     Mean   : 555.197112799
## 3rd Qu.: 585.757530171     3rd Qu.: 14.08033594170     3rd Qu.: 718.0000000000
## Max.   :15261.080946900     Max.   :100.00000298000     Max.   :17271.0000000000
## NA's   :4568              NA's   :4568              NA's   :4
##      tractsnap          year          area
## Min.   : 0.000000000000    Min.   :2019      Length:72531
## 1st Qu.: 67.000000000000    1st Qu.:2019      Class :character
## Median : 152.000000000000    Median :2019      Mode :character
## Mean   : 201.753181574      Mean   :2019
## 3rd Qu.: 282.000000000000    3rd Qu.:2019
## Max.   :2175.000000000000    Max.   :2019

```

```
## NA's :4
## senior_la_n snap_la_n
## Min. : 0.0000000000 Min. : 0.0000000000
## 1st Qu.: 25.8028090326 1st Qu.: 4.59093786055
## Median : 125.9808855350 Median : 24.81192350920
## Mean : 211.4167894130 Mean : 61.94091236500
## 3rd Qu.: 308.0129940800 3rd Qu.: 78.62356775790
## Max. :10348.5798401000 Max. :1582.31071307000
## NA's :29953 NA's :29953
## senior_la_pct snap_la_pct
## Min. : 0.00000000000 Min. : 0.00000000000
## 1st Qu.: 0.66776057096 1st Qu.: 0.30361097156
## Median : 3.15722085022 Median : 1.59971655055
## Mean : 5.08570513756 Mean : 4.06176372883
## 3rd Qu.: 7.53046431934 3rd Qu.: 4.86934977853
## Max. :100.00000000000 Max. :85.09091031990
## NA's :29953 NA's :29918
```

```
colSums(is.na(data2019_clean))
```

```
## censustract state county urban
## 0 0 0 0
## pop2010 ohu2010 laseniors1 laseniors10
## 0 0 19989 64765
## laseniors1share laseniors10share lasnap1share lasnap10share
## 19989 64765 19966 64666
## lasnap1 lasnap10 laseniors20 laseniorshalf
## 19989 64765 71025 4568
## laseniorshalfshare tractseniors tractsnap year
## 4568 4 4 0
## area senior_la_n snap_la_n senior_la_pct
## 0 29953 29953 29953
## snap_la_pct
## 29918
```

```
dim(data2015_clean)
```

```
## [1] 72864 25
```

```
names(data2015_clean)
```

```
## [1] "censustract" "state" "county"
## [4] "urban" "pop2010" "ohu2010"
## [7] "laseniors1" "laseniors10" "lasnap1"
## [10] "laseniors1share" "laseniors10share" "lasnap1share"
## [13] "lasnap10share" "lasnap10" "laseniors20"
## [16] "laseniorshalf" "laseniorshalfshare" "tractseniors"
## [19] "tractsnap" "year" "area"
## [22] "senior_la_n" "snap_la_n" "senior_la_pct"
## [25] "snap_la_pct"
```

```
str(data2015_clean)
```

```
## tibble [72,864 x 25] (S3: tbl_df/tbl/data.frame)
## $ censustract : chr [1:72864] "01001020100" "01001020200" "01001020300" "01001020400" ...
## $ state : chr [1:72864] "Alabama" "Alabama" "Alabama" "Alabama" ...
## $ county : chr [1:72864] "Autauga" "Autauga" "Autauga" "Autauga" ...
```

```
## $ urban          : num [1:72864] 1 1 1 1 1 1 1 0 0 0 ...
## $ pop2010        : num [1:72864] 1912 2170 3373 4386 10766 ...
## $ ohu2010        : num [1:72864] 693 743 1256 1722 4082 ...
## $ laseniors1     : num [1:72864] 162 51 190 237 362 ...
## $ laseniors10    : num [1:72864] 0 0 0 0 0 ...
## $ lasnap1        : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
## $ laseniors1share : num [1:72864] 0.085 0.0235 0.0563 0.0539 0.0336 ...
## $ laseniors10share : num [1:72864] 0 0 0 0 0 ...
## $ lasnap1share    : num [1:72864] 0.1148 0.0561 0.04 0.0142 0.0294 ...
## $ lasnap10share   : num [1:72864] 0 0 0 0 0 ...
## $ lasnap10       : num [1:72864] 0 0 0 0 0 ...
## $ laseniors20    : num [1:72864] 0 0 0 0 0 0 0 0 0 0 ...
## $ laseniorshalf  : num [1:72864] 199 139 346 767 840 ...
## $ laseniorshalfshare : num [1:72864] 0.104 0.0642 0.1026 0.1748 0.078 ...
## $ tractseniors   : num [1:72864] 221 214 439 904 1126 ...
## $ tractsnap      : num [1:72864] 112 202 120 82 488 118 218 115 367 394 ...
## $ year           : num [1:72864] 2015 2015 2015 2015 2015 ...
## $ area           : chr [1:72864] "Urban" "Urban" "Urban" "Urban" ...
## $ senior_la_n    : num [1:72864] 162 51 190 237 362 ...
## $ snap_la_n      : num [1:72864] 79.5 41.7 50.3 24.4 119.9 ...
## $ senior_la_pct  : num [1:72864] 8.5 2.35 5.63 5.39 3.36 ...
## $ snap_la_pct    : num [1:72864] 11.48 5.61 4 1.42 2.94 ...
```

```
summary(data2015_clean)
```

```
## censustract      state      county
## Length:72864     Length:72864   Length:72864
## Class :character  Class :character  Class :character
## Mode :character   Mode :character   Mode :character
##
##
##
##      urban          pop2010      ohu2010
## Min.   :0.00000000000000000000  Min.   : 0.00000000  Min.   : 0.00000000
## 1st Qu.:1.00000000000000000000  1st Qu.: 2883.7500000  1st Qu.: 1102.00000000
## Median :1.00000000000000000000  Median : 4002.0000000  Median : 1521.00000000
## Mean   :0.757191480018          Mean   : 4237.2850516  Mean   : 1601.83756039
## 3rd Qu.:1.00000000000000000000  3rd Qu.: 5323.0000000  3rd Qu.: 2018.00000000
## Max.   :1.00000000000000000000  Max.   :37452.0000000  Max.   :16043.00000000
##      laseniors1      laseniors10      lasnap1
## Min.   : 0.0000000000  Min.   : 0.0000000000  Min.   : 0.0000000000
## 1st Qu.: 0.0000000000  1st Qu.: 0.0000000000  1st Qu.: 0.0000000000
## Median : 111.152876843  Median : 0.0000000000  Median : 20.1084352526
## Mean   : 230.558283934  Mean   : 12.0994974777  Mean   : 71.4784359749
## 3rd Qu.: 390.563446517  3rd Qu.: 0.0000000000  3rd Qu.: 103.0891161450
## Max.   :10225.694634300  Max.   :2530.9999906900  Max.   :1521.2415445500
##      laseniors1share      laseniors10share      lasnap1share
## Min.   :0.00000000000000000000  Min.   :0.00000000000000000000  Min.   :0.00000000000000000000
## 1st Qu.:0.00000000000000000000  1st Qu.:0.00000000000000000000  1st Qu.:0.00000000000000000000
## Median :0.0276110682663          Median :0.00000000000000000000  Median :0.0132343918389
## Mean   :0.0563605601272          Mean   :0.00413024682724          Mean   :0.0460208855680
## 3rd Qu.:0.0988746660144          3rd Qu.:0.00000000000000000000  3rd Qu.:0.0673183309857
## Max.   :1.0000000298000          Max.   :0.57680036250900          Max.   :1.0000000217700
##      lasnap10share      lasnap10
## Min.   :0.00000000000000000000  Min.   : 0.00000000000000000000
```

```
## 1st Qu.:0.000000000000000 1st Qu.: 0.00000000000
## Median :0.000000000000000 Median : 0.00000000000
## Mean :0.00311638858043 Mean : 3.71003338017
## 3rd Qu.:0.000000000000000 3rd Qu.: 0.00000000000
## Max. :0.88888886780400 Max. :1108.68866587000
## laseniors20 laseniorshalf laseniorhalfshare
## Min. : 0.00000000000 Min. : 0.000000000 Min. :0.0000000000000
## 1st Qu.: 0.00000000000 1st Qu.: 153.679179371 1st Qu.:0.0424771214540
## Median : 0.00000000000 Median : 345.999997497 Median :0.0914459654018
## Mean : 1.49147507485 Mean : 395.213484884 Mean :0.0964164394142
## 3rd Qu.: 0.00000000000 3rd Qu.: 567.160194513 3rd Qu.:0.1374729178860
## Max. :2081.12934891000 Max. :15260.141464200 Max. :1.0000000298000
## tractseniors tractsnap year
## Min. : 0.000000000 Min. : 0.000000000 Min. :2015
## 1st Qu.: 318.000000000 1st Qu.: 69.000000000 1st Qu.:2015
## Median : 495.000000000 Median : 158.000000000 Median :2015
## Mean : 552.645805885 Mean : 207.022205753 Mean :2015
## 3rd Qu.: 716.000000000 3rd Qu.: 293.000000000 3rd Qu.:2015
## Max. :17271.000000000 Max. :2152.000000000 Max. :2015
## area senior_la_n snap_la_n
## Length:72864 Min. : 0.00000000000 Min. : 0.0000000000000
## Class :character 1st Qu.: 0.00000000000 1st Qu.: 0.0000000000000
## Mode :character Median : 5.79194930115 Median : 0.933608777821
## Mean : 123.77540303500 Mean : 38.252579180900
## 3rd Qu.: 167.16781171100 3rd Qu.: 37.768104004800
## Max. :10225.69463430000 Max. :1521.241544550000
## senior_la_pct snap_la_pct
## Min. : 0.0000000000000 Min. : 0.0000000000000
## 1st Qu.: 0.0000000000000 1st Qu.: 0.0000000000000
## Median : 0.158221080176 Median : 0.0655059568322
## Mean : 2.961367271190 Mean : 2.4835267010000
## 3rd Qu.: 4.125176027870 3rd Qu.: 2.4183350131700
## Max. :100.000000000000 Max. :100.0000021770000
```

```
colSums(is.na(data2015_clean))
```

```
## censustract state county urban
## 0 0 0 0
## pop2010 ohu2010 laseniors1 laseniors10
## 0 0 0 0
## lasnap1 laseniors1share laseniors10share lasnap1share
## 0 0 0 0
## lasnap10share lasnap10 laseniors20 laseniorshalf
## 0 0 0 0
## laseniorhalfshare tractseniors tractsnap year
## 0 0 0 0
## area senior_la_n snap_la_n senior_la_pct
## 0 0 0 0
## snap_la_pct
## 0
```

```
#summary(data2015_clean$snap_la_rate)
#summary(data2019_clean$snap_la_rate)
```

```

#summary(data2015_clean$senior_la_rate)
#summary(data2019_clean$senior_la_rate)

#data2015_clean %>%
#  summarise(
#    n_over_1 = sum(snap_la_rate > 1, na.rm = TRUE),
#    max_rate = max(snap_la_rate, na.rm = TRUE)
#  )

#data2019_clean %>%
#  summarise(
#    n_over_1 = sum(snap_la_rate > 1, na.rm = TRUE),
#    max_rate = max(snap_la_rate, na.rm = TRUE)
#  )

#data2015_clean %>%
#  filter(snap_la_rate > 1) %>%
#  select(
#    censustract,
#    urban,
#    tractsnap,
#    lasnap1,
#    lasnap10,
#    snap_la_n,
#    snap_la_rate
#  ) %>%
#  head(20)

#data2019_clean %>%
#  filter(snap_la_rate > 1) %>%
#  select(
#    censustract,
#    urban,
#    tractsnap,
#    lasnap1,
#    lasnap10,
#    snap_la_n,
#    snap_la_rate
#  ) %>%
#  head(20)

#data2015_clean %>%
#  summarise(
#    n_numerator_gt_denominator =
#      sum(snap_la_n > tractsnap, na.rm = TRUE)
#  )

#data2019_clean %>%
#  summarise(
#    n_numerator_gt_denominator =
#      sum(snap_la_n > tractsnap, na.rm = TRUE)
#  )

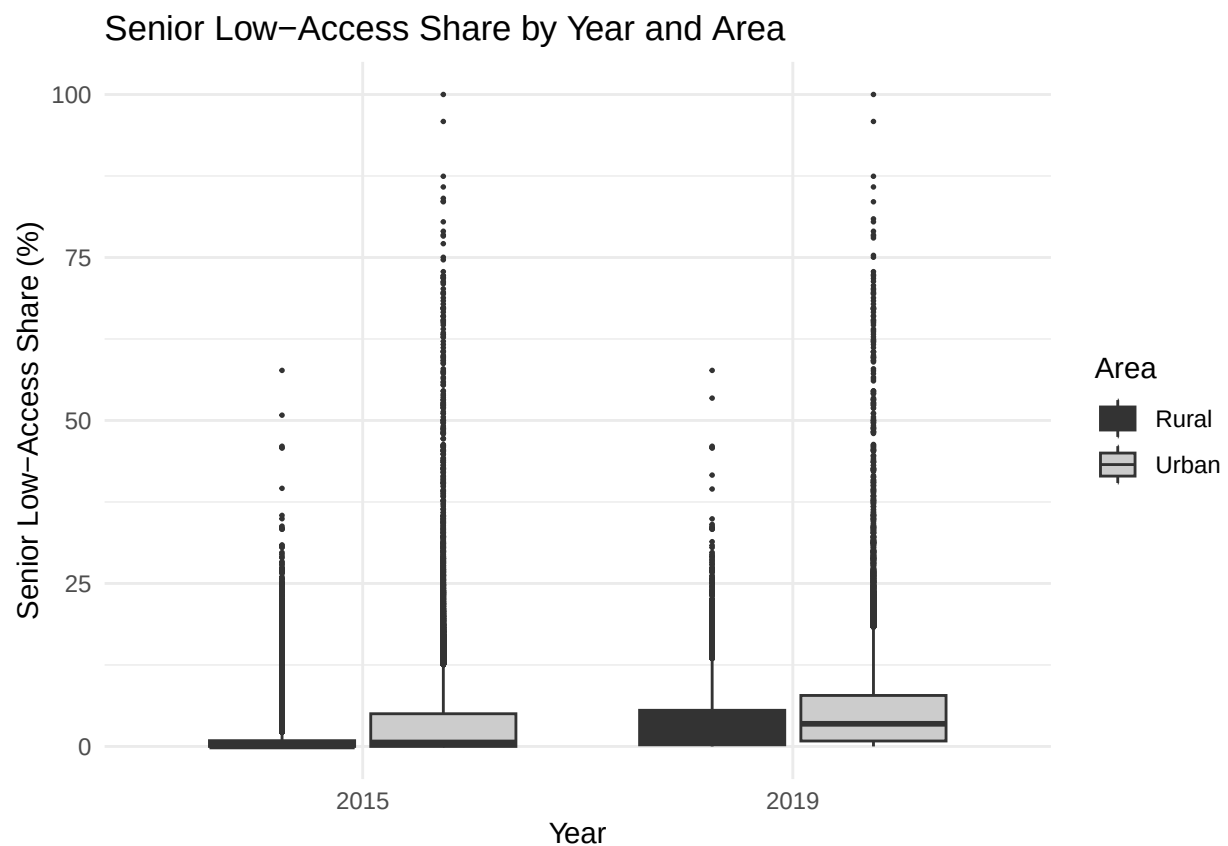
# Senior Boxplot
ggplot(

```

```

combined_data,
aes(
  x = factor(year),
  y = senior_la_pct,
  fill = area
)
) +
geom_boxplot(
  outlier.size = 0.3,
  na.rm = TRUE
) +
scale_fill_grey() +
labs(
  title = "Senior Low-Access Share by Year and Area",
  x = "Year",
  y = "Senior Low-Access Share (%)",
  fill = "Area"
) +
theme_minimal()

```



```

# SNAP boxplot
ggplot(
  combined_data,
  aes(
    x = factor(year),
    y = snap_la_pct,

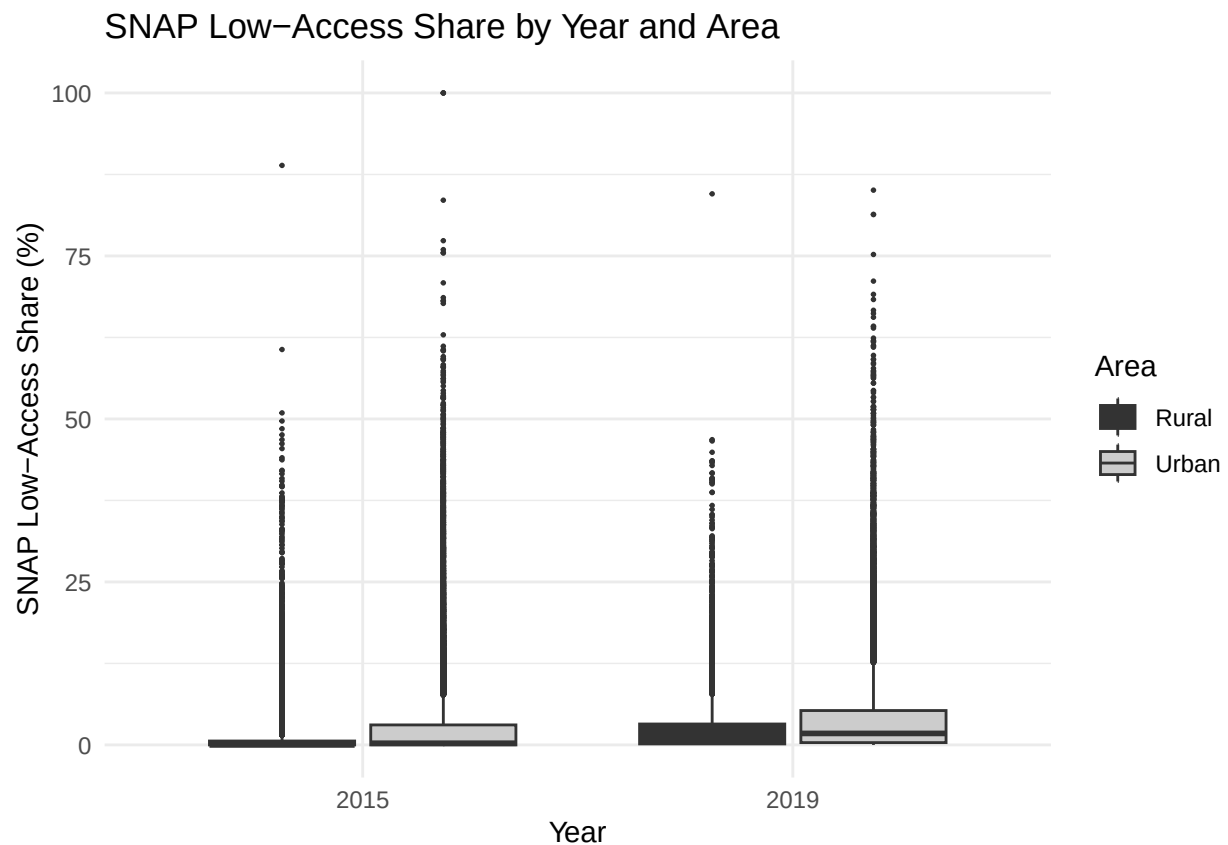
```



```

    fill = area
  )
) +
  geom_boxplot(
    outlier.size = 0.3,
    na.rm = TRUE
  ) +
  scale_fill_grey() +
  labs(
    title = "SNAP Low-Access Share by Year and Area",
    x = "Year",
    y = "SNAP Low-Access Share (%)",
    fill = "Area"
  ) +
  theme_minimal()

```

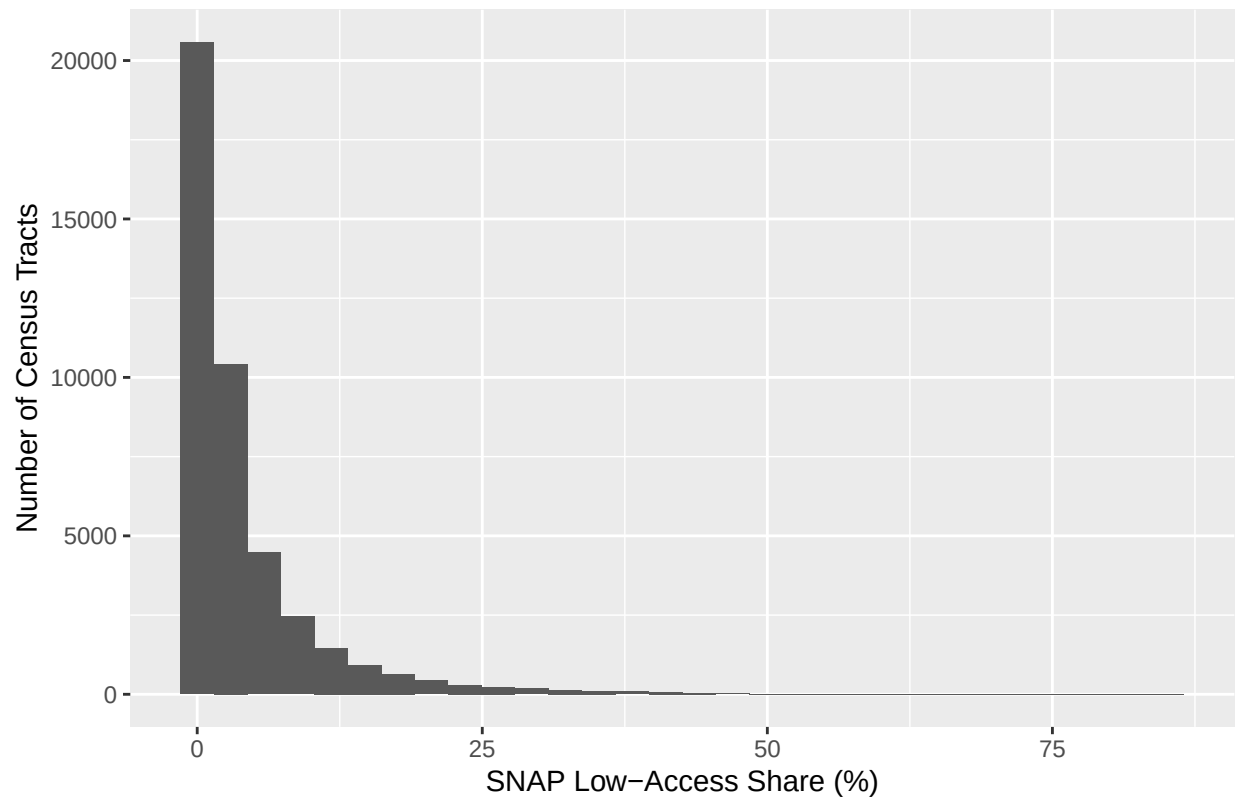


```

library(ggplot2)
ggplot(data2019_clean, aes(x = snap_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of SNAP Low-Access Share, 2019",
    x = "SNAP Low-Access Share (%)",
    y = "Number of Census Tracts"
  )

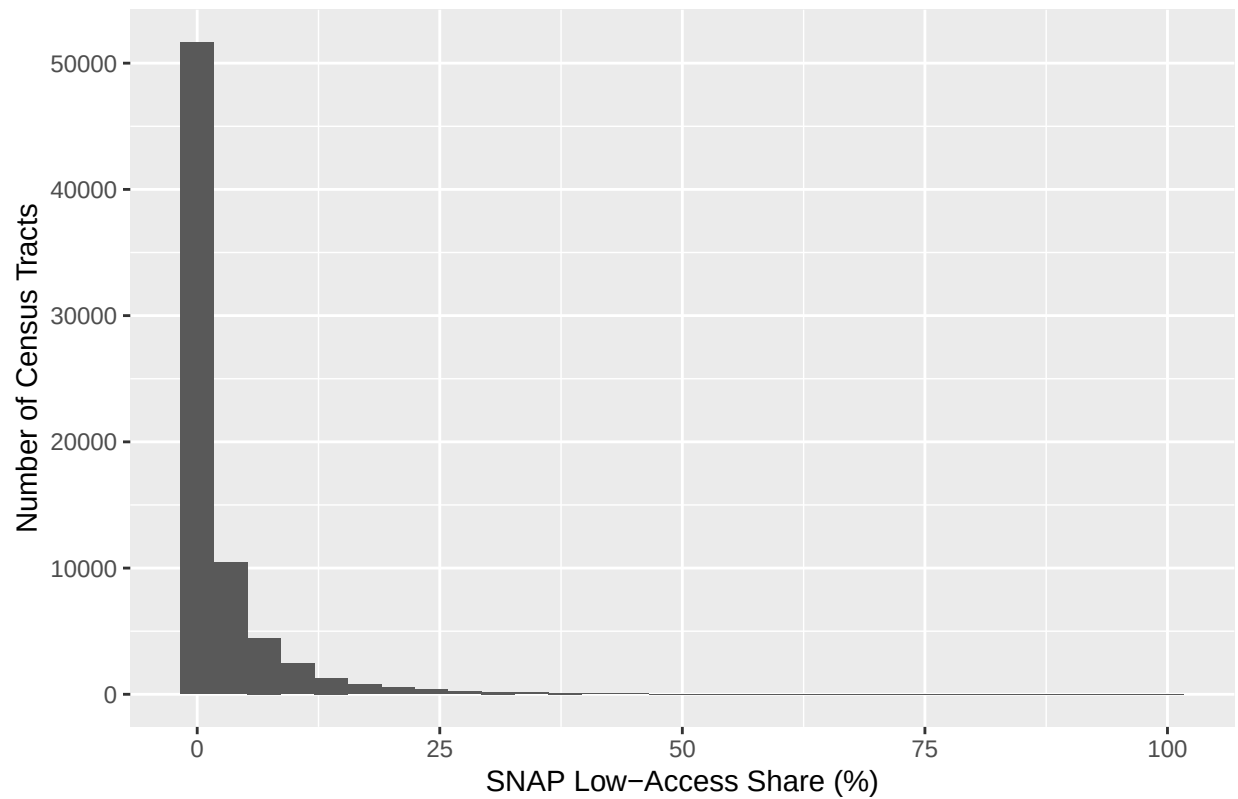
```

Distribution of SNAP Low-Access Share, 2019



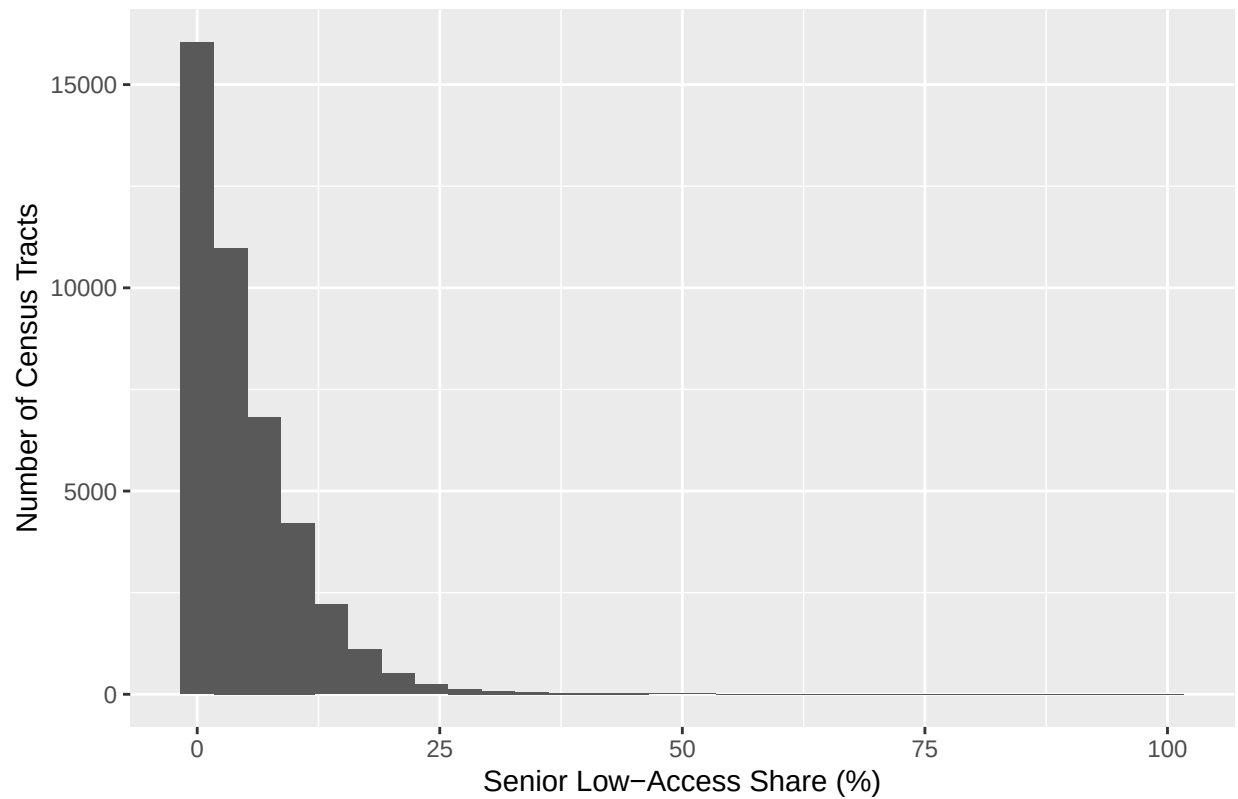
```
library(ggplot2)
ggplot(data2015_clean, aes(x = snap_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of SNAP Low-Access Share, 2015",
    x = "SNAP Low-Access Share (%)",
    y = "Number of Census Tracts"
  )
```

Distribution of SNAP Low-Access Share, 2015



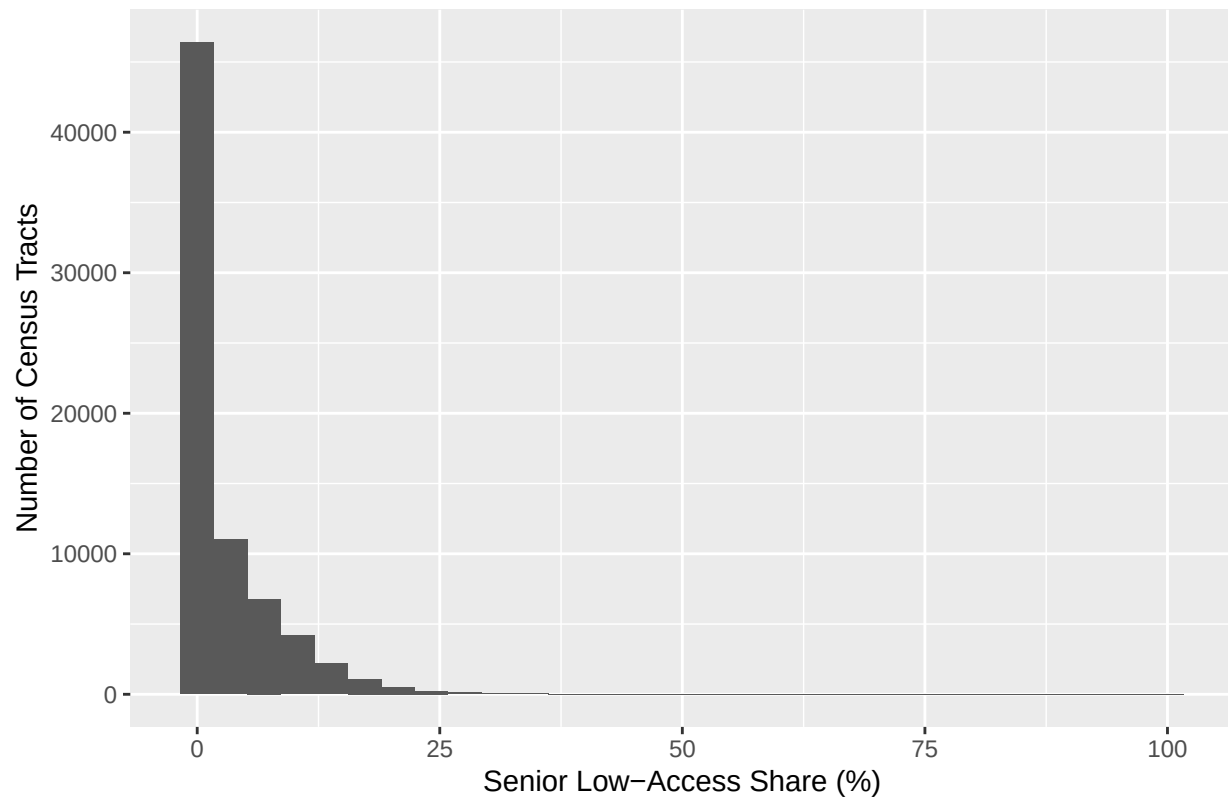
```
library(ggplot2)
ggplot(data2019_clean, aes(x = senior_la_pct)) +
  geom_histogram(bins = 30, na.rm = TRUE) +
  labs(
    title = "Distribution of Senior Low-Access Share, 2019",
    x = "Senior Low-Access Share (%)",
    y = "Number of Census Tracts"
  )
```

Distribution of Senior Low-Access Share, 2019



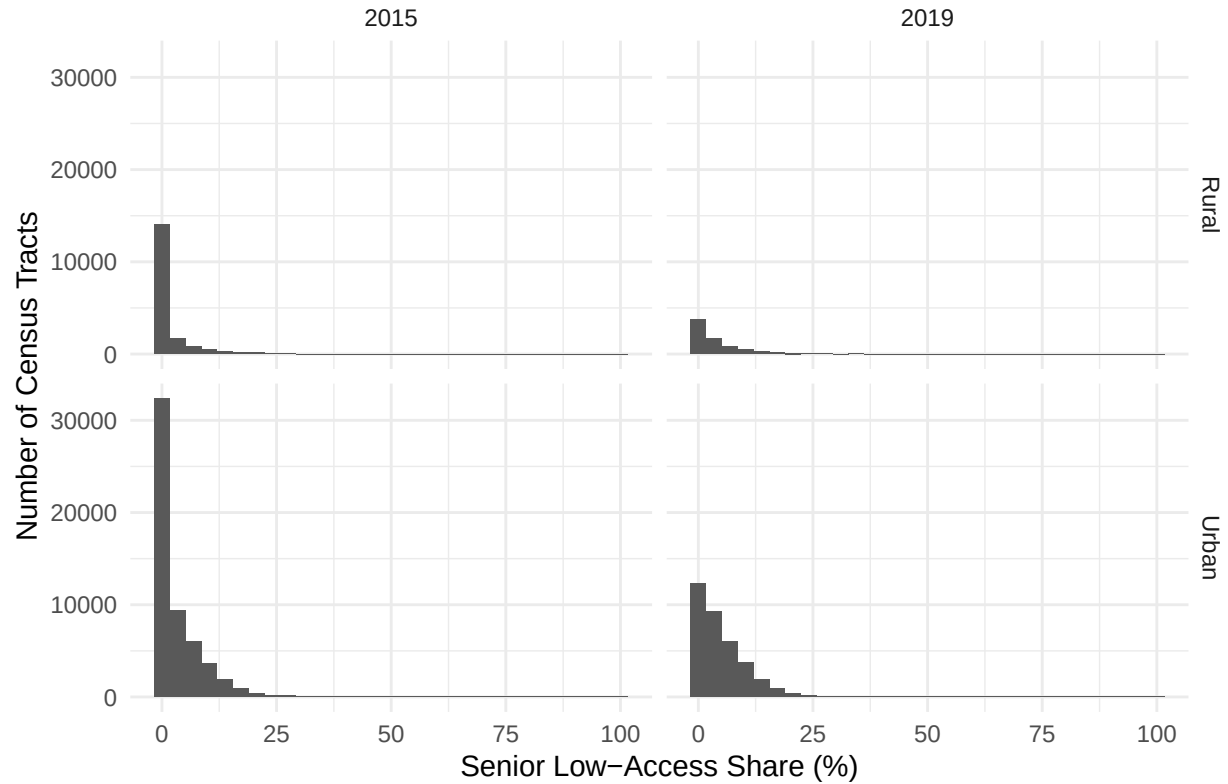
```
ggplot(data2015_clean, aes(x = senior_la_pct)) +  
  geom_histogram(bins = 30, na.rm = TRUE) +  
  labs(  
    title = "Distribution of Senior Low-Access Share, 2015",  
    x = "Senior Low-Access Share (%)",  
    y = "Number of Census Tracts"  
  )
```

Distribution of Senior Low-Access Share, 2015



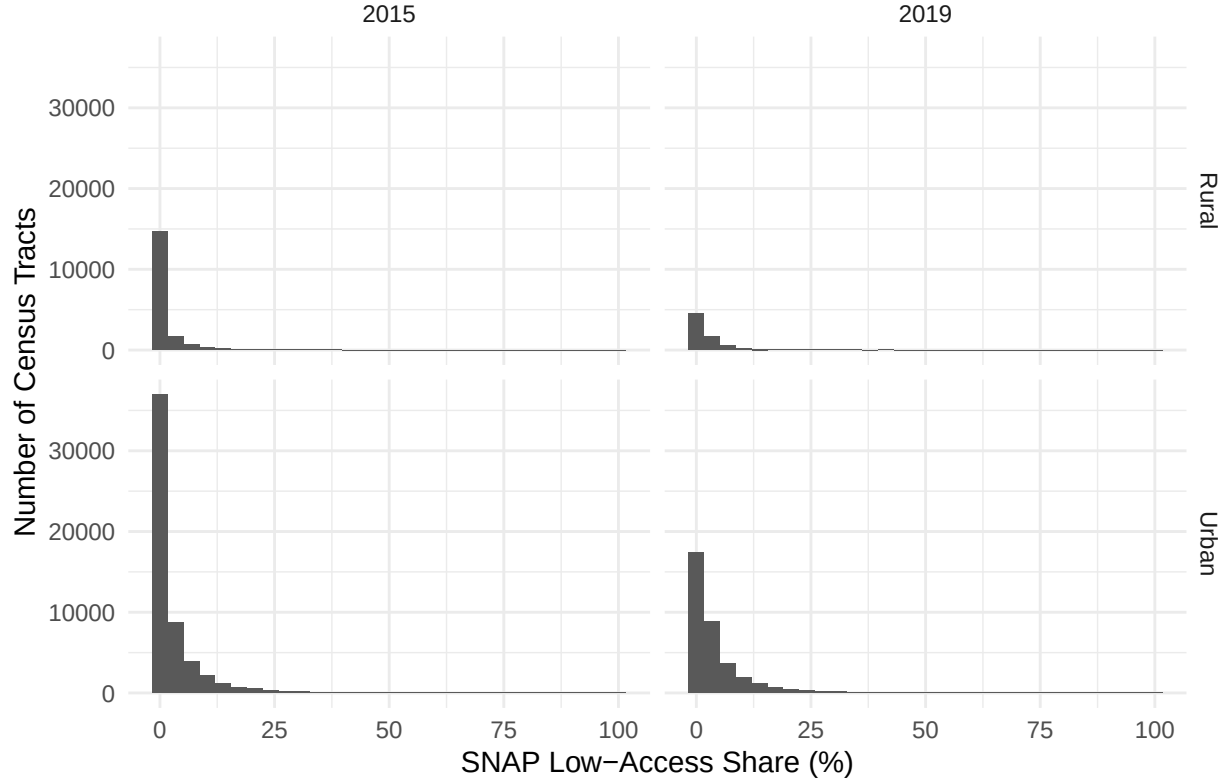
```
ggplot(combined_data, aes(x = senior_la_pct)) +  
  geom_histogram(bins = 30, na.rm = TRUE) +  
  facet_grid(area ~ year) +  
  labs(  
    title = "Distribution of Senior Low-Access Share",  
    x = "Senior Low-Access Share (%)",  
    y = "Number of Census Tracts"  
  ) +  
  theme_minimal()
```

Distribution of Senior Low-Access Share



```
ggplot(combined_data, aes(x = snap_la_pct)) +  
  geom_histogram(bins = 30, na.rm = TRUE) +  
  facet_grid(area ~ year) +  
  labs(  
    title = "Distribution of SNAP Low-Access Share",  
    x = "SNAP Low-Access Share (%)",  
    y = "Number of Census Tracts"  
  ) +  
  theme_minimal()
```

Distribution of SNAP Low-Access Share



Clean Summary Statistics Section

```
summary_table <- combined_data %>%
  group_by(area, year) %>%
  summarise(
    Senior_Mean = round(mean(senior_la_pct, na.rm = TRUE), 1),
    Senior_Median = round(median(senior_la_pct, na.rm = TRUE), 1),
    SNAP_Mean = round(mean(snap_la_pct, na.rm = TRUE), 1),
    SNAP_Median = round(median(snap_la_pct, na.rm = TRUE), 1),
    n_tracts = n(),
    .groups = "drop"
  )

kable(summary_table,
  caption = "Mean and median low-access share (%) for seniors and SNAP households, by area and year",
  col.names = c("Area", "Year", "Senior Mean", "Senior Median",
    "SNAP Mean", "SNAP Median", "N Tracts"))
```

Table 3: Mean and median low-access share (%) for seniors and SNAP households, by area and year.

Area	Year	Senior Mean	Senior Median	SNAP Mean	SNAP Median	N Tracts
Rural	2015	1.7	0.0	1.3	0.0	17692
Rural	2019	4.0	1.7	2.8	1.0	17362
Urban	2015	3.4	0.6	2.9	0.3	55172
Urban	2019	5.3	3.5	4.3	1.8	55169

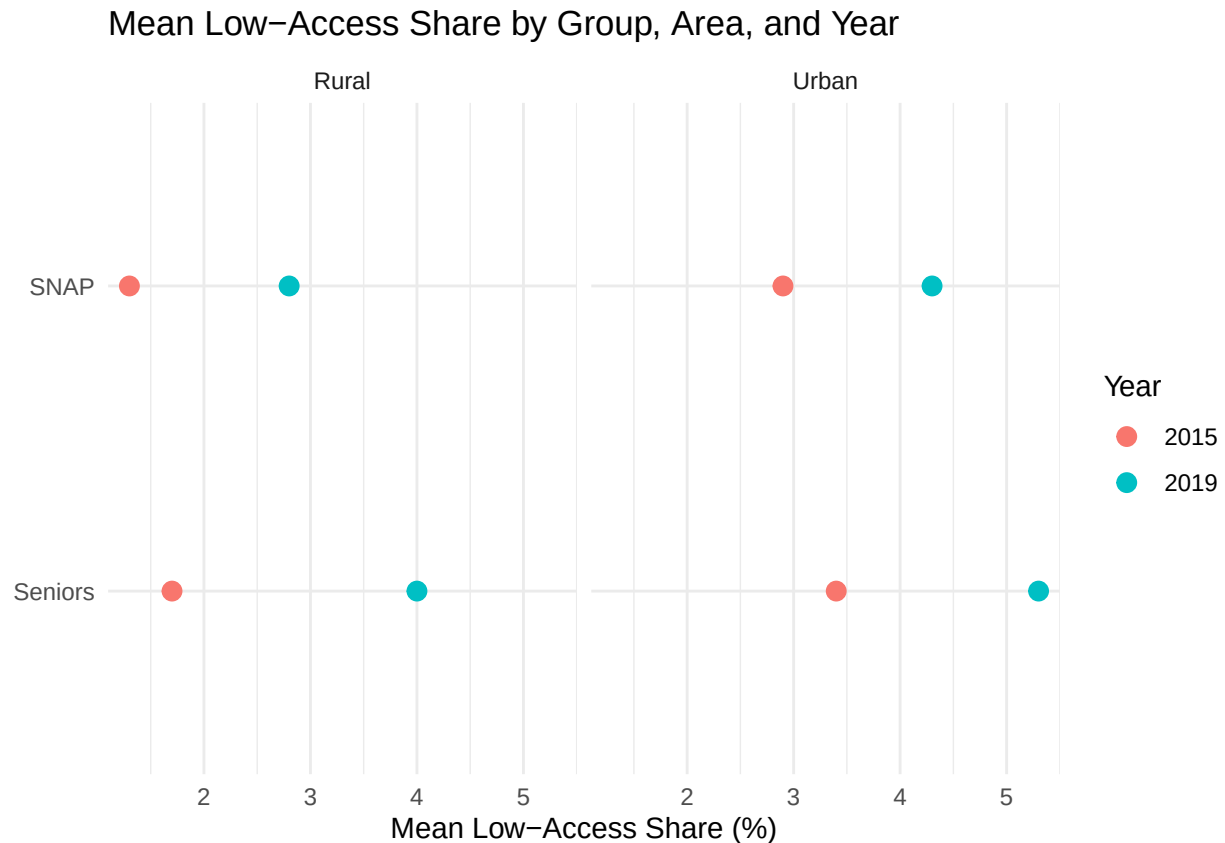
To talk about... median much lower than means = skewed data. Urban vastly outnumber rural. Values increase across all four rows

keeping in mind a rural tract requires households and seniors to be at least 10 miles away compared to a min of 1

```
library(tidyr)

dot_data <- summary_table %>%
  select(area, year, Senior_Mean, SNAP_Mean) %>%
  pivot_longer(cols = c(Senior_Mean, SNAP_Mean),
               names_to = "Group", values_to = "Mean_Rate") %>%
  mutate(Group = recode(Group, Senior_Mean = "Seniors", SNAP_Mean = "SNAP"))

ggplot(dot_data, aes(x = Mean_Rate, y = Group, color = factor(year))) +
  geom_point(size = 3) +
  facet_wrap(~area) +
  labs(
    title = "Mean Low-Access Share by Group, Area, and Year",
    x = "Mean Low-Access Share (%)",
    y = NULL,
    color = "Year"
  ) +
  theme_minimal()
```



Across both Senior and SNAP low-access shares, 2019 shows a higher mean rate than 2015, and Urban tracts show a higher mean rate than Rural tracts, in both years and for both groups

Because “low-access” is defined using a much stricter distance threshold in urban areas (1 mile) than rural areas (10 miles), a higher urban low-access rate does not necessarily mean urban residents travel further to reach a supermarket than rural residents. This may just reflect that urban areas are held to a stricter access standard. A rural resident traveling several miles to a store may still count as “having access” under the rural threshold, while a much shorter distance in a city would count as “low access.”

As a result, the lower mean low-access share in rural areas should not be read as evidence that rural food access is genuinely better. Rural residents may need to travel considerably farther in absolute terms before being classified as low-access at all, meaning the rural rate likely understates the true burden of distance compared to the urban rate. This distinction should be kept in mind whenever urban and rural low-access rates are compared directly in this report, rather than treating them as measuring the same underlying concept.